PROPOSAL FIS-01623

Status: Submitted **Submission-Date:** 07/11/2023, 23:14:30

SECTION A - ADMINISTRATIVE FORM

PRINCIPAL INVESTIGATOR

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Current Role

Current Role:

Permanent Researcher-Technologist III Level

Current Job Location/Home Institution:

INFN Laboratories of Frascati

Residence Data

Country:

ITALIA

Province:

Roma - 058

City:

Roma - H501

Address:

Via delle Benedettine 92

REQUIREMENT

Grant:
Advanced Grant

I declare that I am scientifically independent and active in research for more than 12 years:

YES

HOST INSTITUTION

Type of institution:

Public research entities

Name:

130 -- Istituto Nazionale di Fisica Nucleare

Fiscal Code / Tax ID:

84001850589

VAT Code:

04430461006

Pec:

Host status:

EPR

Street, Number, City, District and Country:

Via Enrico Fermi, 54,Frascati (Roma),RM,ITA

PROJECT DETAILS

Title:

EXOtic atoms’ spectroscopy with ultra-fast TIming CdZnTe detectors

Acronym:

EXOTIC

Grant:

Advanced Grant

Project Duration:

36

Main ERC:

PE2 - Fundamental Constituents of Matter

Other ERC fields 1:

-

Other ERC fields 2:

ERC subfields 1:

PE2_2 - Phenomenology of fundamental interactions
PE2_3 - Experimental particle physics with accelerators
PE2_6 - Nuclear, hadron and heavy ion physics

ERC subfields 2:

ERC subfields 3:

Keywords:

HADRONIC PHYSICS
NUCLEAR PHYSICS
SENSORS

Other Keywords:

Kaon Mass
Fast response CZT detectors
Exotic (Kaonic) atoms

Proposal Abstract:

The EXOTIC (EXOtic atoms' spectroscopy with ultra-fast TIming CdZnTe detectors) project aims at performing unique **high-precision crucial measurements** of intermediate-mass kaonic atoms transitions, fundamental to boost the understanding of the strong interaction in the strangeness sector and to resolve pending "strangeness" puzzles, by developing and using **innovative ultra-fast timing CdZnTe (CZT)** solid-state detector systems.

The main EXOTIC scientific goals are to deliver: a new value of the charged kaon mass with a precision below 5 keV, **solving the 20-years** old 5 sigmas discrepancy between the two most-precise reported values; the energies and widths of selected transitions in KC, KAl, KS and KPb with a precision below a few per mille, and their absolute yields.

EXOTIC will also **deliver advanced X-ray spectrometers** based on new CZT detectors and innovative digital electronics, with energy resolution < 1.2 %, time resolution ~ 1 ns and efficiency > 70% in the 25-350 keV range.

The EXOTIC apparatus will be realized following a modular concept, such as to be easily adapted for use at different kaon beam lines, such as the DAFNE collider of INFN or the K1.8BR beamline at J-PARC.

Preliminary CZT tests at DAFNE, performed with a few cm2 module, **confirmed the feasibility of the EXOTIC ambitious project**, the outcomes of which will have **strong impact** in fundamental science, in particular in the low-energy strangeness QCD where a **lack of reliable data** is preventing from a proper tuning of theoretical models, and in astrophysics (EOS of neutron stars), but also for society, where this cost-effective device can be employed in nuclear medicine and nuclear security. In particular, the high time resolution and the high energy resolution at room temperature will push to explore the use of these detectors for positron emission tomography (PET) applications.

To successfully fulfil the EXOTIC's goals a six people research team (PI plus 5 ad-hoc researchers) will be established at the INFN Laboratories of Frascati (LNF), where the DAFNE colliders resides, and where a **longstanding tradition in exotic atoms physics experiments** will be a great benefit for the EXOTIC team, which will also rely on the expertise and know-how of the LNF's staff, as well as on the possibility to access the BTF facility for the characterization of the detectors.

BUDGET

Expenditure item	Euros
A. Principal Investigator:	0
B.Personnel :	739900
C.Instruments and equipment:	0

D.Other operational costs:	989540
E.Scientific and technical consulting services:	500000
F. Overheads:	147980
Total Cost:	2377420

OTHER FUNDING AND RELATED PI TIME COMMITMENT EXTERNAL TO THE PROJECT

Proposal Title	Funding Source	Amount (Euro)	Start Date	End Date	PI Role	Relation to current Proposal	Time, in percentage terms dedicated to the project
JRA8-Advanced ultra-fast solid STate detectors for high precision RAdiation spectroscopy (ASTRA)	STRONG2020 (H2020)	50000	01/06/2019	31/07/2024			

TEACHING AND MANAGEMENT COMMITMENTS EXTERNAL TO THE PROJECT

Institution	Start Date	End Date	Role	Percentage time dedicated to teaching and management commitments external to the project

SECTION B - SCIENTIFIC PROPOSAL

SYNOPSIS

Main ERC Field:

PE2 - Fundamental Constituents of Matter

Other ERC Field 1:

-

Other ERC Field 2:

ERC Subfield 1:

PE2_2 - Phenomenology of fundamental interactionsPE2_3 - Experimental particle physics with acceleratorsPE2_6 - Nuclear, hadron and heavy ion physics

ERC Subfield 2:

ERC Subfield 3:

PROPOSAL SUMMARY

Understanding the strong interaction in the strangeness sector is still far from being complete, especially in the low-energy regime where the theory is non-perturbative. This is extremely important also to comprehend the chiral symmetry breaking which is at the basis of most part of the visible mass in the Universe. In this context, the kaonic atoms supply the theory with key ingredients, since in the innermost levels of a kaonic atom the kaon resents the strong interaction (at nearly threshold). The measurement of transitions in kaonic atoms is, however, difficult due to technological limitations in radiation detector systems, and the existent database is not only far from being complete, but also not reliable. As a paradoxical situation in this sector, the charged kaon mass itself stands out: the two most precise measurements are 5 sigmas far apart. The EXOTIC project has the ambitious goal to fill in these gaps, to fuel the theory with new and reliable data, to obtain a new precise value of the charged kaon mass and, moreover, to achieve these results by developing a cost-effective detector system which can also be successfully employed for the benefit of society.

Objectives and state of the art

The EXOTIC's main goal is to perform **unique measurements** of X-ray transitions in selected kaonic atoms, at the DAFNE collider of INFN Laboratories of Frascati (LNF) and/or at the K1.8BR beamline at J-PARC, by developing a **groundbreaking** CdZnTe (CZT) X-ray detectors system with **unprecedented timing and spectroscopic capabilities** in the 25-350 keV range, with high linearity and stability in an accelerator environment.

EXOTIC will deliver:

- A new reliable value of the charged kaon mass with a precision below 5 keV.
- The energies, widths and yields of a series of X-ray transitions in KC, KAl, KS, and KPb, with unprecedented precisions of a few per mille and, for many of them, also as first-ever measurements.
- An innovative X-ray spectrometer prototype based on new CZT detectors and digital electronics with energy resolution < 1.2 %, time resolution < 1 ns and detection efficiency > 70% in the 25-350 keV range.

These kaonic atom measurements will contribute **to overcoming the present impasse** in the knowledge of the interaction mechanisms between kaons and nuclei, schematized in Fig. S1.

Presently, the charged kaon mass precision is 13 keV, resulting from an average of several measurements where the error has been artificially increased by a scale factor S=2.5 to take into account the large disagreement (5 sigmas) between the two most recent and precise results [1] (see Fig. S2).

The present database of kaonic atom transitions was populated in the 1970s and 1980s [2] (see Fig. S3), and it is the basis of many models and theories of the strong interaction in systems with strangeness at low-energies [3].
However, the most recent measurements, in part performed by the PI, revealed how **the older data are not reliable,** casting doubts on the full set of existing data [3]. Furthermore, many of the earliest measurements of the strong interaction-induced shifts and widths suffer from **exceptionally large uncertainties** and, in some cases, they are **in contrast** with the most recent theoretical models [4]**. Are the models wrong or are the experimental data incorrect?**

New precise measurements are then **crucial and strongly demanded by various scientific communities** [5] [6], and DAFNE and K1.8BR represent unique facilities to perform them.

From the technological point of view, the development of the new X-ray detection systems will exploit the expertise and the consolidated collaboration with the research group of the University of Palermo (CZT detectors and electronics). The research activity will be focused on **overcoming the performance of the current state of art of CZT-spectrometers**, aiming to obtain: (a) excellent energy resolution at room temperature (≤ 1.2 % FWHM at 60 keV; ≤ 1% FWHM at 122 keV; ≤ 1% FWHM at 350 keV), (b) high detection efficiency (> 70 % up to 350 keV), (c) high time resolution (≤ 1 ns; key for PET systems) and (d) time stability. The **advancement of knowledge** brought by the EXOTIC project will establish a scientific and technological base for the design and development of new-generation X-ray spectrometer systems.

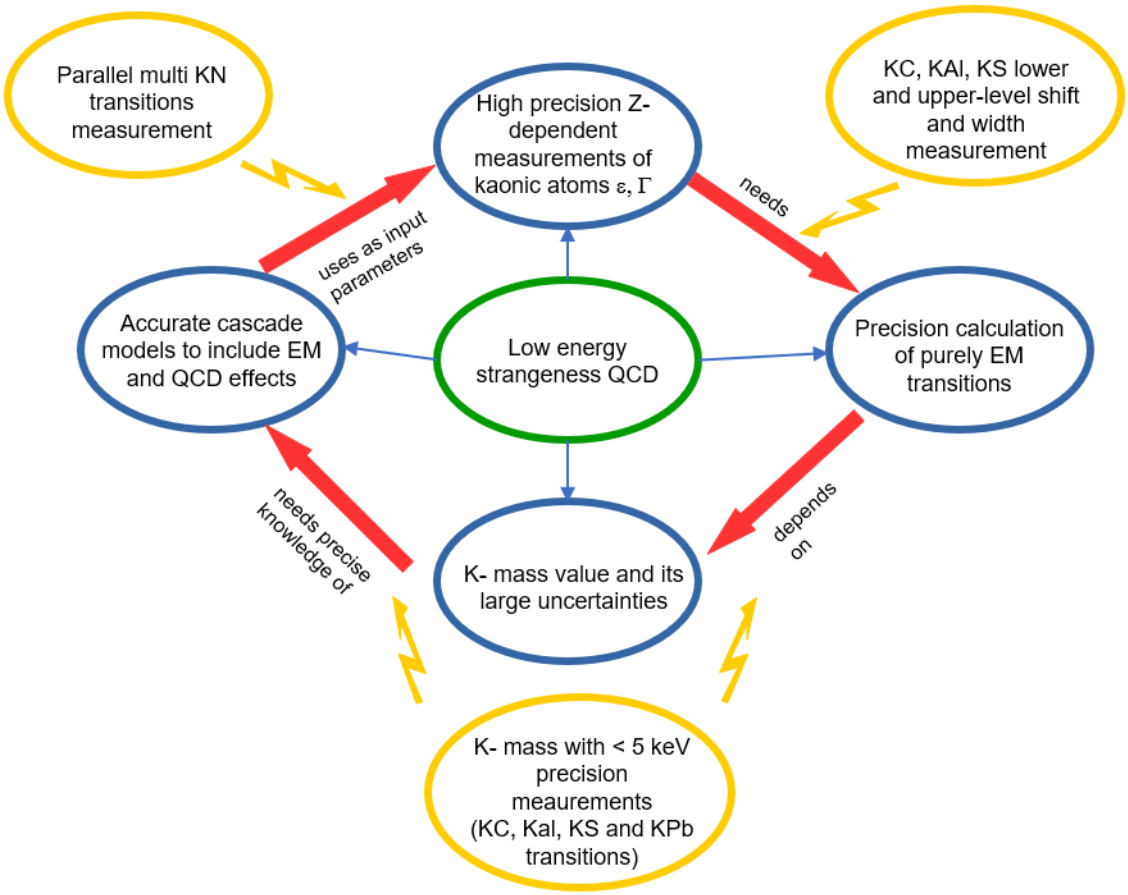


Fig. S1. The impact of EXOTIC in low-energy strangeness QCD.

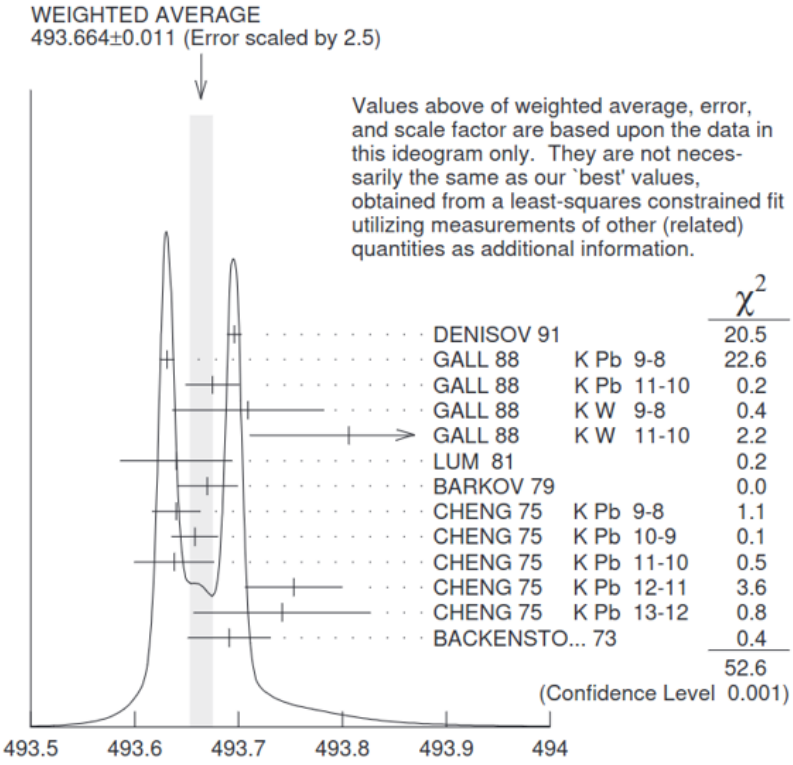


Fig. S2: value of the weighted average K^- mass published in the PDG (https://pdg.lbl.gov/2020/listings/rpp2020-list-K-plus-minus.pdf)

Table 1
Compilation of K^- atomic data

Nucleus	Transition	ϵ (keV)	Γ (keV)	Y	Γ_u (eV)
He	3→2	−0.04 ±0.03	–	–	–
		−0.035±0.012	0.03 ±0.03	–	–
Li	3→2	0.002±0.026	0.055±0.029	0.95±0.30	–
Be	3→2	−0.079±0.021	0.172±0.58	0.25±0.09	0.04±0.02
^{10}B	3→2	−0.208±0.035	0.810±0.100	–	–
^{11}B	3→2	−0.167±0.035	0.700±0.080	–	–
C	3→2	−0.590±0.080	1.730±0.150	0.07±0.013	0.99±0.20
O	4→3	−0.025±0.018	0.017±0.014	–	–
Mg	4→3	−0.027±0.015	0.214±0.015	0.78±0.06	0.08±0.03
Al	4→3	−0.130±0.050	0.490±0.160	–	–
		−0.076±0.014	0.442±0.022	0.55±0.03	0.30±0.04
Si	4→3	−0.240±0.050	0.810±0.120	–	–
		−0.130±0.015	0.800±0.033	0.49±0.03	0.53±0.06
P	4→3	−0.330±0.08	1.440±0.120	0.26±0.03	1.89±0.30
S	4→3	−0.550±0.06	2.330±0.200	0.22±0.02	3.10±0.36
		−0.43 ±0.12	2.310±0.170	–	–
		−0.462±0.054	1.96 ±0.17	0.23±0.03	2.9 ±0.5
Cl	4→3	−0.770±0.40	3.80 ±1.0	0.16±0.04	5.8 ±1.7
		−0.94 ±0.40	3.92 ±0.99	–	–
		−1.08 ±0.22	2.79 ±0.25	–	–
Co	5→4	−0.099±0.106	0.64 ±0.25	–	–
Ni	5→4	−0.180±0.070	0.59 ±0.21	0.30±0.08	5.9 ±2.3
		−0.246±0.052	1.23 ±0.14	–	–
Cu	5→4	−0.240±0.220	1.650±0.72	0.29±0.11	7.0 ±3.8
		−0.377±0.048	1.35 ±0.17	0.36±0.05	5.1 ±1.1
Ag	6→5	−0.18 ±0.12	1.54 ±0.58	0.51±0.16	7.3 ±4.7
Cd	6→5	−0.40 ±0.10	2.01 ±0.44	0.57±0.11	6.2 ±2.8
In	6→5	−0.53 ±0.15	2.38 ±0.57	0.44±0.08	11.4 ±3.7
Sn	6→5	−0.41 ±0.18	3.18 ±0.64	0.39±0.07	15.1 ±4.4
Ho	7→6	−0.30 ±0.13	2.14 ±0.31	–	–
Yb	7→6	−0.12 ±0.10	2.39 ±0.30	–	–
Ta	7→6	−0.27 ±0.50	3.76 ±1.15	–	–
Pb	8→7	–	0.37 ±0.15	0.79±0.08	4.1 ±2.0
		−0.020±0.012	–	–	–
U	8→7	−0.26 ±0.4	1.50 ±0.75	0.35±0.12	45 ±24

Fig. S3: Summary of the existing measurements of lower transitions in kaonic atoms

Setups and expected results

The EXOTIC project will realize a **prototype of an innovative experimental apparatus based** on the new CZT technology, depicted in Fig. S4, to be easily and readily used both in collider and extracted beam experiments.

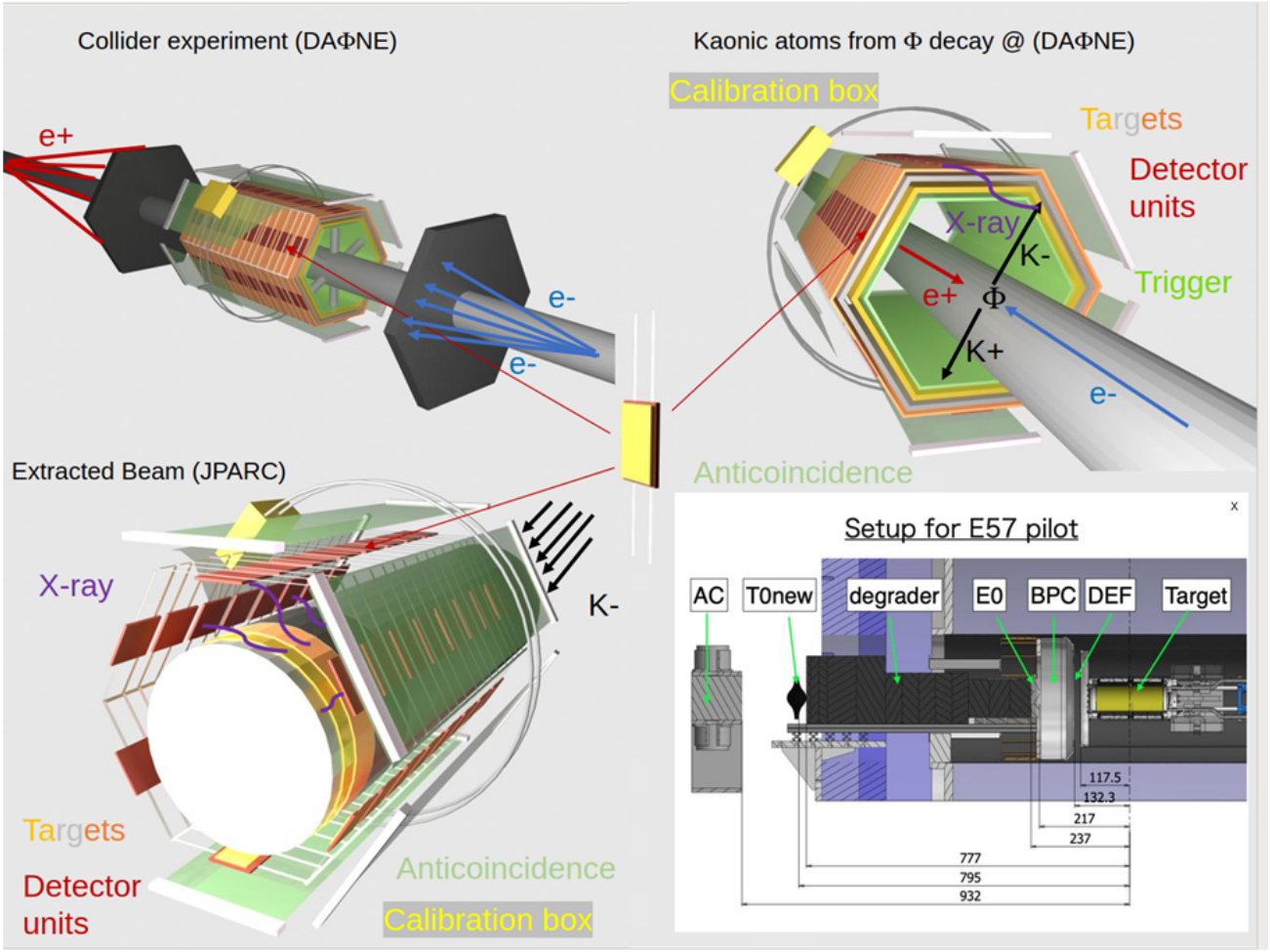


Fig. S4: (not in scale) sketch of the possible EXOTIC apparatuses to be used at DAΦNE (up) and JPARC (down). In the centre, the single unit is shown. More details in the text.

EXOTIC will have a **modular concept** based on a single unit, formed by a 1 cm2 coplanar-grid CZT detector coupled to its preamplifier board. It could **easily switch** between DAFNE and K1.8BR using different structures holding the units, through which a temperature stabilization around 15-20°C could be achieved by means of a water-cooling system.

At DAFNE, back-to-back low momentum K+K- pairs will be first detected by a layer of scintillators (trigger), then the K- will form kaonic atoms in the C, Al, S and Pb targets producing X-rays, then detected by the CZT units. A further external scintillator layer will serve as anticoincidence to reduce the background, already shielded by two Pb structures placed at the two ends of the Interaction Region (IR).

At J-PARC, the setup will be included in the E57 experiment [7], exploiting the ~ 10^8/s K- extracted from the main proton beam. Almost 90% of these K- will stop in the carbon degrader of the experiment, which will be modified with the insertion of a series of targets with an equivalent density to the original one. An external scintillator layer will be used as anticoincidence while a series of existing detectors will act as trigger and beam monitor [7].

In both cases, **a crucial role** will be played by the ~ 1 ns time resolution of the CZT detectors, to greatly enhance the Signal-to-Background ratio.

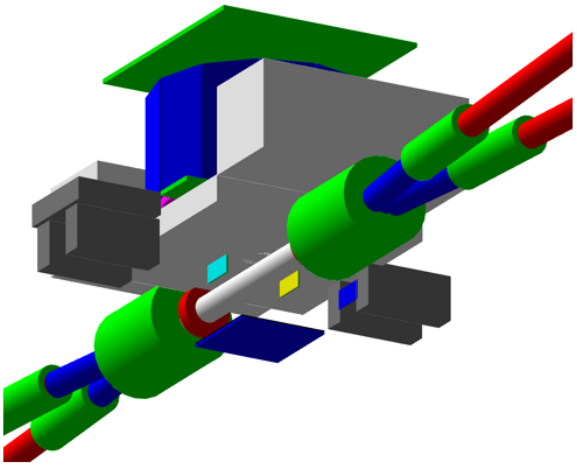
The CZT units will be placed on hexagonal prism structures, with each face having 50 detectors for a total active area of 300 cm2.

Periodical calibrations of the detectors will be performed with radioactive sources, like 241Am, enclosed in a remotely on-off switchable shielding box and placed on a motorized circular rail around the hexagonal prism to shine all six faces. **This will ensure a drastic reduction of systematics**.

From preliminary MC simulations performed including a 15,6 cm2 CZT in the SIDDHARTA-2 code, the experiment presently running at DAFNE, **expected results on the main kaonic transitions of interest can be inferred in true beam conditions** and are reported in Fig. S5, while a sketch of the simulated geometry is visible in Fig. S6 and, as an example, in Fig. S7 the spectrum obtainable with a 10pb-1 run with an Al+C target is shown.

Transition	Energy (keV)	events / 10 pb^{-1} with 15,6 cm^2	Yield / K^- stop	Expect events in 60 days with 300 cm^2	Natural Linewidth Γ (keV)	CdZnTe resolution σ (keV)	Convoluted resolution σ (keV)	estimated error on the energy (keV) for S/B=1/5	estimated error on the width Γ (keV) for S/B=1/5	estimated error on the energy (keV) for S/B=1/50	estimated error on the width Γ (keV) for S/B=1/50
KC(3-->2)	63	1065	0,07	28386	1,8	1,26	1,475	0,020	0,039	0,062	0,124
KAl(4-->3)	106	967	0,55	202512	0,45	2,12	2,129	0,011	0,021	0,033	0,067
KS(4-->3)	161	940	0,22	78743	2,2	3,22	3,353	0,027	0,053	0,084	0,169
KC(4-->3)	22	943	0,07	25135	0	0,44	0,440	0,006	0,012	0,020	0,039
KC(5-->3)	32	912	0,035	12154	0	0,64	0,640	0,013	0,026	0,041	0,082
KAl(5-->4)	49	944	0,55	197695	0	0,98	0,980	0,005	0,010	0,016	0,031
KAl(6-->4)	76	968	0,27	99518	0	1,52	1,520	0,011	0,022	0,034	0,068
KAl(6-->5)	27	933	0,55	195392	0	0,54	0,540	0,003	0,005	0,009	0,017
KAl(7-->5)	43	900	0,27	92527	0	0,86	0,860	0,006	0,013	0,020	0,040
KS(5-->4)	74	1590	0,22	133193	0	1,48	1,480	0,009	0,018	0,029	0,057
KS(5-->3)	235	560	0,11	23455	0	4,7	4,700	0,069	0,137	0,217	0,434
KS(6-->5)	40	1350	0,22	113088	0	0,8	0,800	0,005	0,011	0,017	0,034
KS(6-->4)	115	1310	0,11	54869	0	2,3	2,300	0,022	0,044	0,069	0,139
KS(7-->5)	65	1700	0,11	71204	0	1,3	1,300	0,011	0,022	0,034	0,069
KPb(9-->8)	289	170	0,79	51137	0	5,78	5,780	0,057	0,114	0,181	0,361
KPb(10-->9)	207	190	0,4	28938	0	4,14	4,140	0,054	0,109	0,172	0,344
KPb(11-->10)	153	370	0,79	111299	0	3,06	3,060	0,021	0,041	0,065	0,130
KPb(12-->10)	269	130	0,4	19800	0	5,38	5,380	0,085	0,171	0,270	0,541
KPb(11-->9)	359	80	0,79	24065	0	7,18	7,180	0,103	0,207	0,327	0,655
KPb(12-->11)	116	360	0,79	108291	0	2,32	2,320	0,016	0,032	0,050	0,100
KPb(13-->11)	207	270	0,4	41123	0	4,14	4,140	0,046	0,091	0,144	0,289

Fig. S5: Error estimation from the KC and KAl MC simulations. The expected number of events in 60 days assumes a very conservative integrated luminosity of 5 pb^{-1} /day. The yields for the lower transitions are taken from the kaonic atoms' database [6], while for the upper ones no experimental data are available. For the ΔN=1 (green) upper-level transitions the yields are taken, conservatively, equal to the lowest ones even if higher n transitions usually have higher yields. In case of ΔN=2 (red), a further factor 2 less is assumed. Another assumption, based on experimental observations, is that the natural linewidths of the upper levels are order of magnitude smaller than those of the lower levels and, in our case, the final resolution is totally dominated by the detector resolution. Finally, the errors are estimated from the peak resolution and the number of events assuming a highly conservative scenario of a signal-to-background ratio of 1/5, as well as for a 'worst case' scenario of a signal-to-background ratio of 1/50.



15,6 cm^2 CdZnTe

DAΦNE Quadrupoles
SIDDHARTA-2 Luminometer

Targets (attached to Luminometer):
70 μm Al
100 μm C
100 μm S

Distance from IR:
216 mm

Fig. S6: Details of the GEANT4 simulations of 15,6 cm2 CdZnTe inserted in the main SIDDHARTA-2 code.

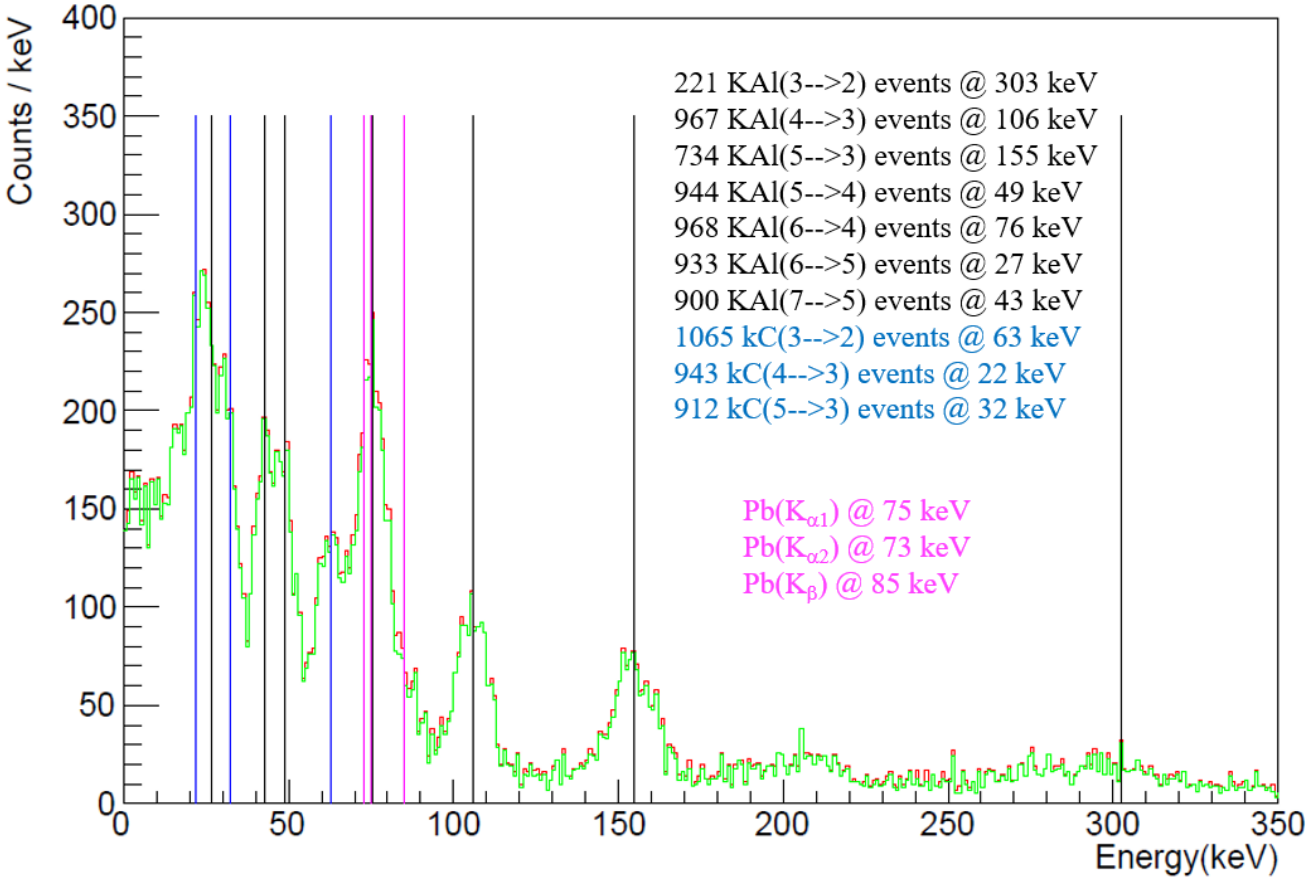


Fig. S7: GEANT4 MC simulations, within the SIDDHARTA-2 geometry, of a 15,6 cm^2 CdZnTe detector placed at 216 mm from the Interaction Region. The statistics corresponds to an integrated luminosity of 10 pb^{-1} (2 days of data taking). The kaonic lines are generated in the target assuming a 100% yield per K^- stop, while the pink lines indicate X-ray fluorescence peaks from the Pb shielding activated by particles lost.

Using the calculations from [8] [9], a function relating each transition energy to the Kaon mass can be obtained; by performing a simultaneous fitting of all the measured transitions in which the various peak positions are parametrized as a function of the Kaon mass, left as a free parameter, this latter will be extracted with a precision below 5 keV.

These numbers show how the main goals of EXOTIC could be already achieved with a DAFNE data-taking of 60 days, corresponding to 300 pb^-1 integrated luminosity; in this case, additional targets (O, Si, Co, Cu, etc.) will be used at J-PARC to consolidate the results and add to the kaonic atoms' database.

The spokespersons of E57 strongly support the EXOTIC project and agreed to collaborate in its integration with the experimental apparatus (see Fig. S10).

The EXOTIC Workplan

I will build up and coordinate a research team composed of myself (70%) and 5 other III Level Researchers (from R1 to R5).

The EXOTIC Workplan is organized in 6 Work Packages (WP), with 3-7 tasks each (T), summarized in the GANTT chart of Fig. S8 with their Main Responsible (MR) indicated with a color-referenced filled cell. The 5 Milestones (M) are indicated with light blue bars at their month of delivery. A detailed description of the individual WP can be found in the budget justification section of part C.

The hiring procedures will start immediately after the notification of the EXOTIC project's approval such as to allow for starting all the tasks in time. In the unwanted case of some delay, the PI will be responsible for the first phases of each task.

In Fig. 9, the PERT diagram is shown, where the task of each member of the team is highlighted in color-referenced frames, while the whole diagram is embedded in the 'PI Supervision' frame.

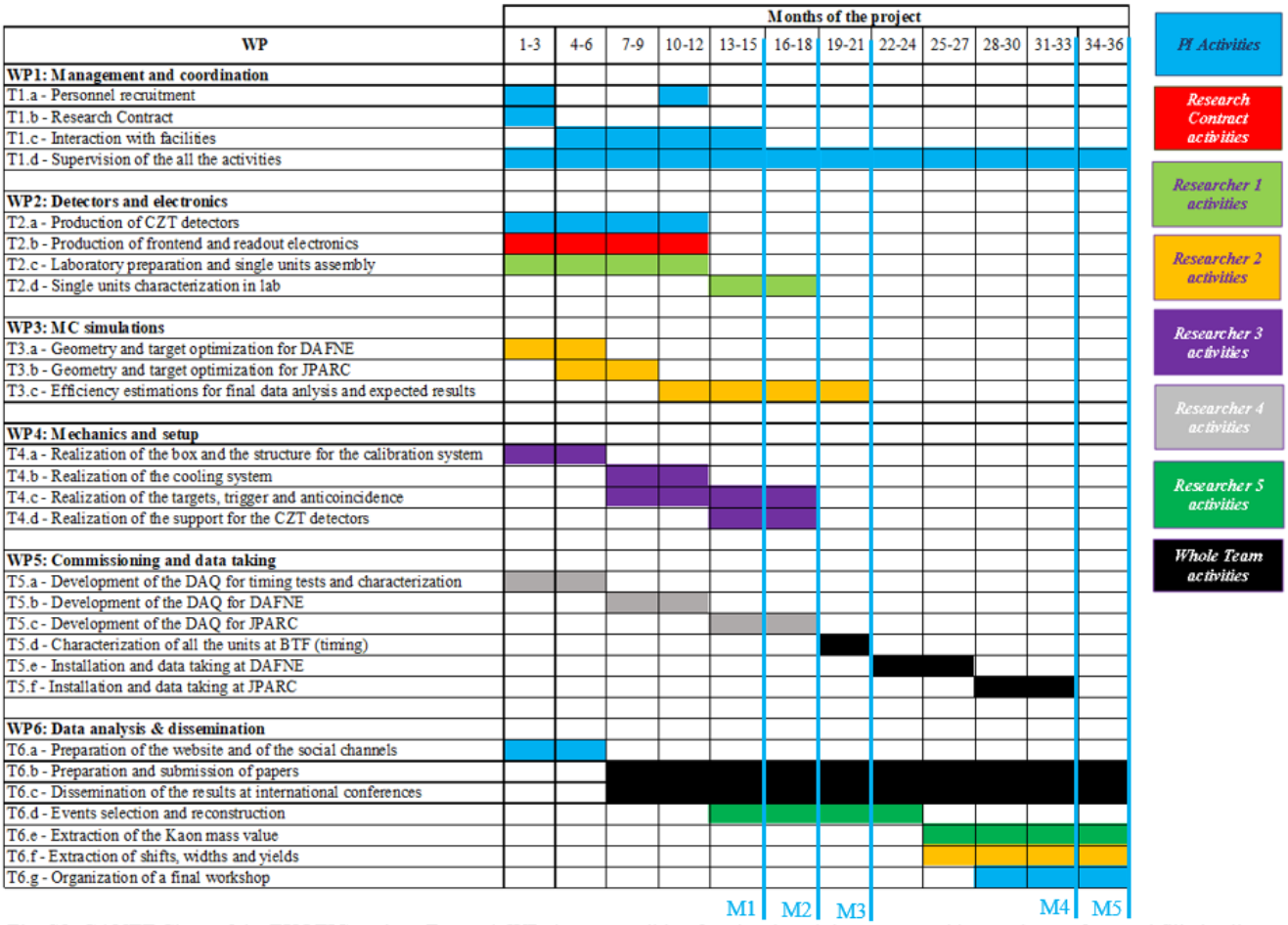


Fig. S8: GANTT Chart of the EXOTIC project. For each WP, the responsible of each sub-task is represented by a colour-referenced filled cell. In light blue, the 5 Milestone are reported at their month of delivery.

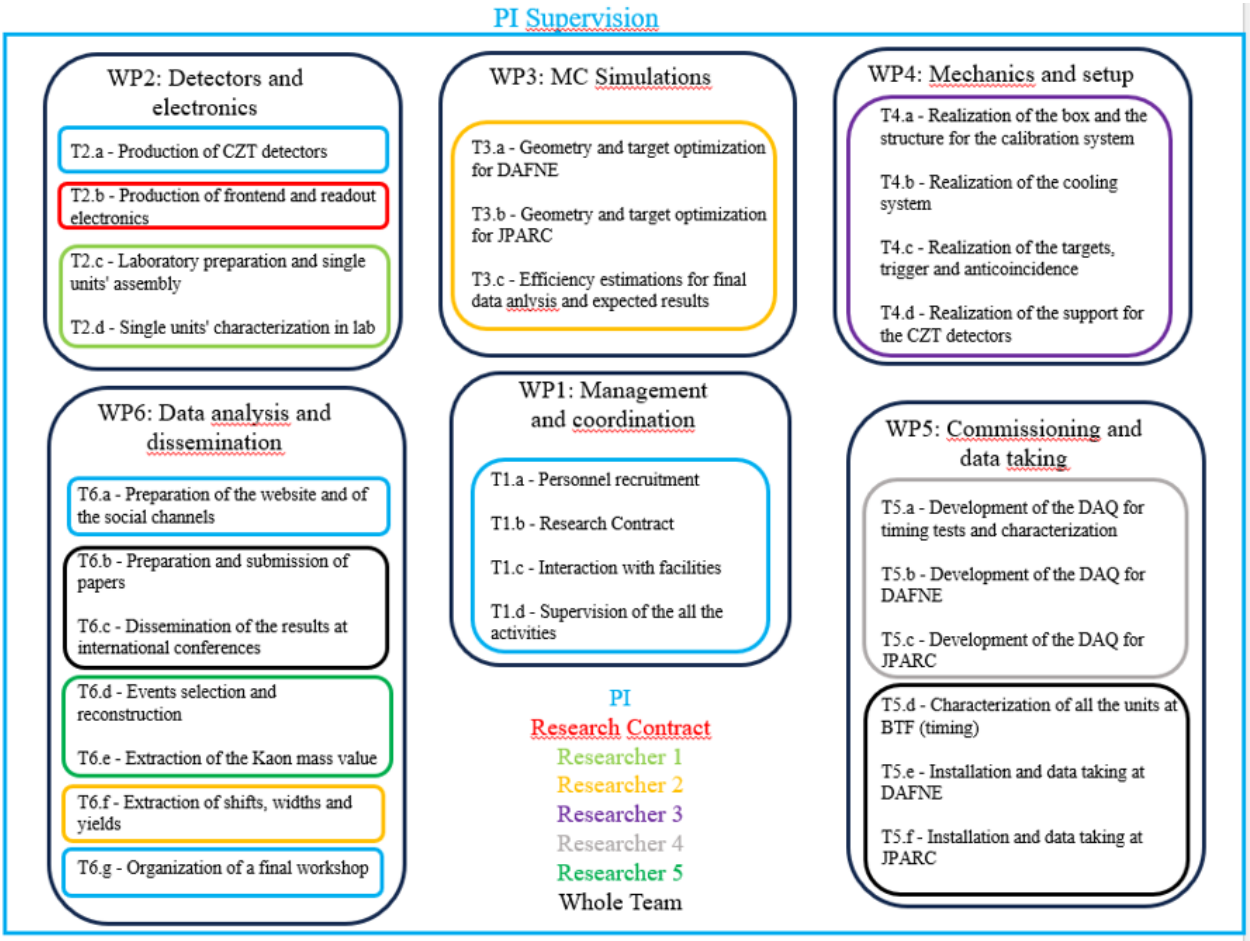
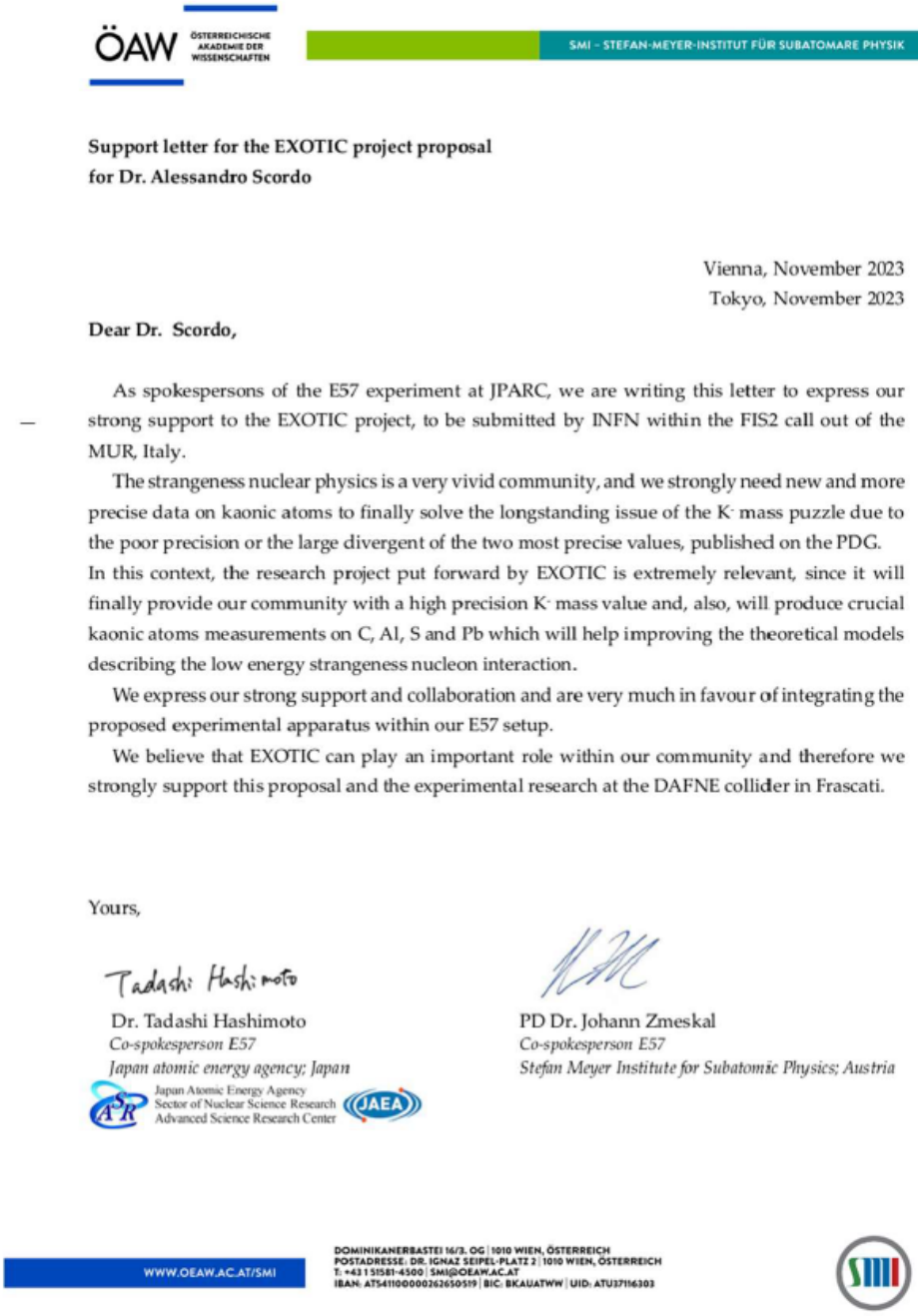


Fig. S9: PERT diagram of EXOTIC. The task of each member of the team are highlighted in color-referenced frames, while the whole diagram is embedded in the 'PI Supervision' frame.



ÖAW ÖSTERREICHISCHE AKADEMIE DER WISSENSCHAFTEN

SMI – STEFAN-MEYER-INSTITUT FÜR SUBATOMARE PHYSIK

Support letter for the EXOTIC project proposal
for Dr. Alessandro Scordo

Vienna, November 2023
Tokyo, November 2023

Dear Dr. Scordo,

As spokespersons of the E57 experiment at JPARC, we are writing this letter to express our strong support to the EXOTIC project, to be submitted by INFN within the FIS2 call out of the MUR, Italy.

The strangeness nuclear physics is a very vivid community, and we strongly need new and more precise data on kaonic atoms to finally solve the longstanding issue of the K^- mass puzzle due to the poor precision or the large divergent of the two most precise values, published on the PDG.
In this context, the research project put forward by EXOTIC is extremely relevant, since it will finally provide our community with a high precision K^- mass value and, also, will produce crucial kaonic atoms measurements on C, Al, S and Pb which will help improving the theoretical models describing the low energy strangeness nucleon interaction.

We express our strong support and collaboration and are very much in favour of integrating the proposed experimental apparatus within our E57 setup.

We believe that EXOTIC can play an important role within our community and therefore we strongly support this proposal and the experimental research at the DAFNE collider in Frascati.

Yours,

Tadashi Hashimoto

Dr. Tadashi Hashimoto
Co-spokesperson E57
Japan atomic energy agency; Japan
Japan Atomic Energy Agency
Sector of Nuclear Science Research
Advanced Science Research Center

PD Dr. Johann Zmeskal
Co-spokesperson E57
Stefan Meyer Institute for Subatomic Physics; Austria

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Fig. S10: Support letter from the E57 spokespersons

Feasibility and validation: the EXOTIC roadmap

EXOTIC proposes a **challenging experiment**, whose feasibility **will be ensured by several factors**:

My know-how and expertise, as PI. EXOTIC will rely on the expertise and know-how I gained not only in the physics of kaonic atoms but also in the large variety of experimental apparatuses I contributed to realise. I already had several occasions to work with VETO systems, scintillators, shielding structures, X-ray detectors and their calibration. I also learned how to build and lead a research team thanks to the three winning projects I coordinated as PI starting in 2016. Finally, as **LNF coordinator of the STRONG-2020 JRA8-ASTRA project,** I was already involved in tests and characterisation of CZT detectors [10] [11] [12], and I got in contact with the world's leading teams in this sector. **All my experience and know-how, together with my capability and experience as a project team leader, make me the ideal PI for the EXOTIC project.**

- *My project coordination expertise,* gained from the **three competitive projects I coordinated as PI**, focused on the Von Hamos X-ray Bragg spectrometer that I started developing in 2016.

- *The participation of the research group of the University of Palermo.* Their long-standing experience with CZT detectors is a **well-grounded guarantee of the success of EXOTIC**.

- *The possibility to perform "on beam" characterization of the detectors at BTF.* This INFN facility is not only an added value but a **crucial key point for the success of the experiment**, guaranteeing an immediate test of the timing response of the detectors.

- *The preliminary tests and MC simulations* performed within SIDDHARTA-2 show **very promising results** for the implementation of CZT detectors in kaonic atoms experiments [10] [11] [12].

All my experience and know-how, together with my capability and experience as a project team leader, make me the ideal PI for the EXOTIC project.

Dissemination and outreach

For the whole duration of the project, a strong effort will be dedicated to the dissemination of the results and the outreach, both in terms of the production of scientific and technological papers and general public dissemination.

As a PI, **I strongly believe in the importance of training young researchers**, thus special attention will be paid to their participation in international schools and conferences to strengthen their knowledge and to have fruitful discussions with international expert researchers.

Our present knowledge on low energy strangeness QCD is still far to be completed, and this is the right time to perform big steps forward exploiting the two best available facilities in the world for such type of measurements.

[1] doi={10.1103/PhysRevD.98.030001}

[2] doi={10.1016/0375-9474(94)90921-0}

[3] doi={10.1103/RevModPhys.91.025006}

[4] doi={10.1103/PhysRevC.106.065201}

[5] https://agenda.infn.it/event/25725/contributions/131346/attachments/79609/103625/Eli-Friedman-talk.pdf

[6] https://agenda.infn.it/event/25725/contributions/131349/attachments/79624/103640/foto-Jarka.pdf

[7] https://indico.ectstar.eu/event/182/contributions/4281/

[8] doi={10.1088/0022-3700/20/4/007}

[9] doi={10.1103/PhysRevA.71.032501}

[10] doi={10.1140/epjs/s11734-023-00881-x}

[11] doi={10.3390/s23177328}

[12] doi={10.48550/arXiv.2310.15062}

CURRICULUM VITAE

Personal Information:

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Date of birth: 26/12/1982
Nationality: Italian

Education:

2011- Ph.D. in Physics
Faculty of Physics, University of Rome “Tor Vergata”, Italy
2006- Master
Faculty of Physics, University of Rome “La Sapienza”, Italy

Current Position(s):

2021- Researcher, staff
INFN Laboratories of Frascati (LNF), Italy

Previous Positions:

2019-2021 Researcher, LNF
2011-2019 PostDoc fellow at LNF

Fellowships and awards:

2021- Scientific Responsible for the MITIQO project at LNF, winner of the competitive call “Gruppi di Ricerca 2020” by Regione Lazio, with a total budget of 150 k€ for the 2022-2024 period. Role: Co-PI, Scientific Responsible.

2019- Winner of the competitive Research for Innovation 2020 (R4I-2020) call of the INFN National Tech Transfer Commission (CNTT) with the TASTE project at LNF. Total budget of 30 k€ financed by the CNTT for the 2020 year. Role: PI.

2015- Winner of the competitive Young Researchers Grant n. 17367/2016 of the CSN5-INFN commission with the VOXES project at LNF. Total budget of 230 k€ for the 2016-2018 biennium. Role: PI.

SUPERVISION OF GRADUATE STUDENTS; PhD POSTDOC STUDENTS:

2023- External supervisor of a master’s degree thesis at Università degli Studi di Roma “La Sapienza”, on CZT detectors’ tests at DAFNE

2020- External supervisor of a master’s degree thesis at Università degli Studi di Roma “La Sapienza”, in the framework of the SIDDHARTA-2 experiment.

2017- External supervisor of a master’s degree thesis at Università degli Studi di Roma “Tor Vergata” in the framework of the VOXES project.

TEACHING ACTIVITIES:

ORGANISATION OF SCIENTIFIC MEETINGS:

2017- Organizer, “Strange and non-strange mesons induced processes studies at DAFNE, JPARC and RIKEN: present and future”, Italy
2017- Organizer, “New frontiers in testing quantum mechanics from underground to the space”, Italy
2018- Organizer, “Is Quantum Theory exact? The quest for the spin-statistics connection violation and related items”, Italy
2018- Organizer, “New Quantum Horizons: from foundations to biology” (LNF, 15-16/11/2018).
2018- Chair, “High Precision X-ray Measurements”, Italy
2018- Organizer, “Low-energy strangeness studies at DAFNE and J-PARC”, Italy
2019- Organizer, “Is Quantum Theory exact? From quantum foundations to quantum applications”, Italy
2021- Chair, “High Precision X-ray Measurements 2021”, Online
2021- Organizer “Fundamental Physics at the strangeness frontier at DAFNE”, Online

2021- Chair, “Fundamental physics with exotic atoms and radiation detectors”, Online
2022- Chair, “High precision measurements of kaonic atoms”, Online
2022- Organizer, “The Hitchhiker’s Advanced Guide to Quantum Collapse Models and their impact in science, phylosophy, technology and biology”, Italy
2023- Organizer, “Applications of radiation detection techniques in fundamental physics, food control, medicine and biology”, Italy
2023- Chair, “High Precision X-ray Measurements 2023”, Italy

2023- Chair, “ROCKSTAR: Towards a ROadmap of the Crucial measurements of Key observables in Strangeness reactions for neutron sTARs equation of state”, Italy

INSTITUTIONAL RESPONSIBILITIES:

2022- INFN coordinator of the WP26: JRA8 Advanced ultra-fast solid-STate detectors for high precision Radiation spectroscopy (ASTRA) within the European Horizon 2020 STRONG-2020 project.

2022- Convener of the “gamma/X-ray detectors” section of the “Nuclear Physics Mid Plan Term in Italy – LNF Session”, Italy

2022- Co-PI and Scientific Responsible for the MITIQO project of Regione Lazio.

2020- PI, INFN local and national responsible for the TASTE project in the INFN-CNTT.

2016 – PI, INFN local and national responsible for the VOXES project in the INFN-CSN5.

2016- Local Committee Member, “INFN2016”, Laboratori Nazionali di Frascati INFN, Italy

REVIEWING ACTIVITIES:

2022 – Review Editor for the “Radiation Detectors and Imaging” and “Materials for detectors” sections of Frontiers in Physics
2022 – Guest Associate Editor in the “Nuclear Physics” section of Frontiers in Physics.
2019 – Editorial Board Member for the Condensed Matter Journal (MDPI).

Peer reviewer for the following journals:
Acta Physica Polonica A
Applied Sciences
Chinese Optics Letters
Computer Physics Communications
Condensed Matter
Crystals
Electronics
Journal of Analytical Atomic Spectrometry
Measurement Science and Technology
Nanomaterials
Nuclear Instruments and Methods in Physics Research Section A
Sensors
Sustainability
Symmetry

MEMBERSHIPS OF SCIENTIFIC SOCIETIES:

MAJOR COLLABORATIONS:

Dr. C. Curceanu, Laboratori Nazionali di Frascati INFN, spokesperson of the SIDDHARTA-2 experiment: Kaonic atoms measurements
Prof. L. Abbene, Università degli Studi di Palermo, Italy: CdZnTe detectors and electronics development.
Prof. J. Zmeskal, SMI, Austria, Spokesperson of STRONG2020 JRA8-ASTRA: CdZnTe detectors’ development.
Dr. A. Zappettini, Director of IMEM-CNR, Italy: CdZnTe detectors’ development.
Dr. T. Hashimoto, Spokesperson of the E57 experiment, JAEA-JPARC, Japan
Prof. P Indelicato, Research Director at Laboratoire Kastler Brossel (CNRS), France: exotic atoms calculations

TRACK RECORDS

Track Record:

(Selection of) 10 publications as lead author:

1) High resolution multielement XRF spectroscopy of extended and diffused sources with a graphite mosaic crystal based Von Hamos spectrometer.
https://doi.org/10.1039/C9JA00269C
JAAS, 2020, 35, 155 – 168

2) Measurements of the strong-interaction widths of the kaonic 3He and 4He 2p levels
https://doi.org/10.1016/j.physletb.2012.06.071
Physics Letters B, Volume 714, Issue 1, 24 July 2012, Pages 40-43
3) VOXES: a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2–20 keV range
https://doi.org/10.1088/1748-0221/13/04/C04002
JINST 13 (2018) no.04, C04002.
4) Strong absorption of hadrons with hidden and open strangeness in nuclear matter
https://doi.org/10.1103/PhysRevLett.123.022002
Phys. Rev. Lett. 123 (2019) 2, 022002
5) Pyrolitic Graphite Mosaic Crystal Thickness and Mosaicity Optimization for an Extended Source Von Hamos X-ray Spectrometer
https://doi.org/10.3390/condmat4020038
Condens.Mat. 4 (2019) no.2, 38.
6) Efficiency measurements and simulations of a HAPG based Von Hamos spectrometer for large sources
https://doi.org/10.1039/D1JA00214G
J. Anal. At. Spectrom., 2021,36, 2485-2491
7) Potentialities of CdZnTe Quasi-Hemispherical Detectors for Hard X-ray Spectroscopy of Kaonic Atoms at the DAFNE Collider
http://doi.org/10.3390/s23177328
Sensors, 2023, Vol. 23, Pages 7328
8) A new kaonic helium measurement in gas by SIDDHARTINO at the DAΦNE collider
https://doi.org/10.1088/1361-6471/ac5dac
J. Phys. G: Nucl. Part. Phys. 49 055106
9) Characterization of a scintillating fibers read by MPPC detectors trigger prototype for the AMADEUS experiment
https://doi.org/10.1088/1748-0221/8/05/T05006
JINST 8 (2013) T05006.
10) New opportunities for kaonic atoms measurements from CdZnTe detectors
DOI: 10.1140/epjs/s11734-023-00881-x
EPJ ST 232 (2023) 1487-1492

(Selection of) 10 Invited talks

1) 12-14/12/2014 Invited Talk at the conference “Strangeness Nuclear Physics 2014”, Changsha (China): “Advances and perspectives in the low-energy kaon-nucleon/nuclei interaction studies at the DAFNE collider” .
2) 02-07/06/2016 Invited plenary talk at the “Meson 2016” conference, Krakow (Poland): “Investigation of the low-energy kaons hadronic interactions in light nuclei by AMADEUS”
3) 04-09/06/2017 Invited talk at the “2nd Jagiellonian Symposium on Fundamental and Applied Subatomic Physics”, Krakow: “VOXES, a new high resolution X-ray spectrometer for low yield measurements in high background environments”
4) 10-15/09/2017 Invited talk at the “EXA 2017” conference, Vienna (Austria): “The Kaonic Atoms research program at DAFNE: from SIDDHARTA to SIDDHARTA-2”
5) 04/10/2017 Invited talk at the “TNPI2017” conference, Cortona: “Investigating low energy strangeness QCD through kaon nucleon interactions by the AMADEUS collaboration”
6) 23-27/10/2017 Invited talk at the “ASTRA” Workshop, Trieste ECT*: Development of a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2-20 keV range: the VOXES experiment.”
7) 21-25/10/2019 Invited Talk at the “STRANEX: Recent progress and perspectives in STRANge EXotic atoms studies and related topics” Workshop, ECT*, Trieste: “X and gamma ray precision measurements for fundamental physics and radioprotection.”
8) 10-15/07/2022, Invited Talk at the “4th Jagiellonian Symposium on Advances in Particle Physics and Medicine” conference, Krakow: “Kaonic atoms at DAFNE: where we are and where we go?”
9) 17-21/10/2022, Invited talk at the “EXOTICO: EXOTIc atoms meet nuclear Collisions for a new frontier precision era in low-energy strangeness physics”, ECT*, Trento, “Radiation detectors for future kaonic atoms measurements at DAFNE“
10) 08/06/2023, Invited Talk at the “Channeling 2023” conference, Riccione (Italy): “X-ray detectors and measurements at LNF for nuclear and fundamental physics, Quantum Gravity, and agrifood applications”

Coordination of Research Projects

1) 2015: PI and Scientific Responsible for the INFN project “VOXES (high resolution Von hamos X-ray spectrometer using HAPG for Extended Sources in a broad energy range)“. Young Researchers Grant n. 17367/2016 of the CSN5-INFN. Total budget of 230 k€, 110 k€ financed by the CSN5-INFN and for 120 k€ by the partner institutions for the 2016-2018 biennium
2) 2021: Scientific Responsible for the Regione Lazio project “MITIQO project (Monitoraggio In situ di Tossicità, Indicazione geografica e Qualità di Olio d’oliva, vino e altri liquidi edibili)”, call “Gruppi di Ricerca 2020” by Regione Lazio. Total budget of 150 k€.
3) 2022: INFN coordinator, within the European STRONG-2020 EXCELLENT SCIENCE - Research Infrastructures project, of the WP26: JRA8 Advanced ultra-fast solid STate detectors for high precision Radiation spectroscopy (ASTRA). Total budget 54k€.

Organization of International Conferences

1) 2022: Convener of the “gamma/X-ray detectors” section of the “Nuclear Physics Mid Plan Term in Italy – LNF Session”
2) 2023: Chair of the “ROCKSTAR: Towards a ROadmap of the Crucial measurements of Key observables in Strangeness reactions for neutron sTARs equation of state”
3) 2018,2021,2023: Chair of the “High Precision X-ray Measurements” conference series.

Description of main research results:

During the last 10 years I have been member of several collaborations working in kaonic atoms (SIDDHARTA-2, E17, E57, E62), Kaon-Nucleon interaction (E15, E31, E45, E72, E80, HADES), Fundamental physics (VIP,VIP-Lead,VIP-2) and X-ray spectroscopy (VOXES, TASTE, MITIQO) experiments.
I am the author or co-author of more than 260 publications (20 as first author, 18 as corresponding author) with more than 2000 citations (iNSPIRE), and I have been invited for 18 talks and seminars at international conferences and institutes/universities.
My main research activities are focused on kaonic atoms physics, where I have the largest number of publication and citations, and X-ray spectroscopy where I started my original research line building several research teams. I was a Winner of the competitive call “Young Researcher Grant 2015” of the INFN CSN5 with the VOXES project at LNF, with a 230 k€ total budget and 6 people research team.
As a PI, national and local responsible of VOXES, I have been responsible for the whole management of the experiment, from the budget’s use to the research team leading, and to the connections with external suppliers. On this occasion, I strengthened my independence as a researcher and my project and team leader skills, with published papers in important journals.
My multidisciplinary interests brought me to be funded with a grant from the Research for Innovation 2020 (R4I-2020) competitive call of the INFN National Tech Transfer Commission (CNTT) with the TASTE project at LNF, with 30 k€ budget and a 4 members research team. Presently, I am also Scientific Coordinator of the MITIQO project, winner of the competitive call “Gruppi di Ricerca 2020” by Regione Lazio, with a total budget of 150 k€ and 7-8 people team. These activities in X-ray detection systems allowed me to start the organization of the “High Precision X-ray Measurements” series of workshops and conferences.
My ability to manage projects as a team leader allowed me to become the INFN group responsible for the STRONG-2020 JRA8-ASTRA project, aiming at developing beyond state-of-art advanced radiation detector systems based on CdZnTe.
Within the activity of JRA8-ASTRA, I got in contact with the world’s main experts in CdZnTe detectors’ design and production, setting the ground for future applications and proposals like EXOTIC.

DESCRIPTION OF ACTIVITIES CONDUCTED DURING THE PhD

Project Title:	institution:	Start Date:	End Date:	Activity description:	PI Role:

SECTION C

RESUME GENERAL INFORMATION

Project Title:

EXOtic atoms’ spectroscopy with ultra-fast TIming CdZnTe detectors

Acronym:

EXOTIC

Principal Investigator Name:

ALESSANDRO SCORDO

Name of the project's host institution:

130 -- Istituto Nazionale di Fisica Nucleare

Duration of the project - X month:

36

State of the art and objectives:

The challenging aim of the EXOTIC (EXOtic atoms' spectroscopy with ultra-fast TIming CdZnTe detectors) project is to address crucial shortcomings in the strangeness sector of the low energy nuclear physics by performing ultra-fast timing high energy resolution kaonic atoms X-ray spectroscopy measurements. To this end, a prototype of a novel X-ray detection system based on modular CdZnTe (CZT) detectors will be realized and used, providing a new fundamental tool for exotic (not only kaonic) atoms research, with a strong potential impact in other fundamental physics sectors and with a broad range of possible societal applications.

State of art: kaonic atoms

The measurement of the radiative transitions in exotic atoms is **the most reliable experimental method to provide fundamental model-independent input to the low-energy QCD** models, which must be tuned with experimental data to describe processes such as the chiral symmetry breaking, providing mass to the (visible) Universe.

In the strangeness sector, particularly of interest because the s quark mass is in between the light and heavy ones, a lack of data is preventing a proper tuning of the models and, consequently, progress (see Fig. SA1).

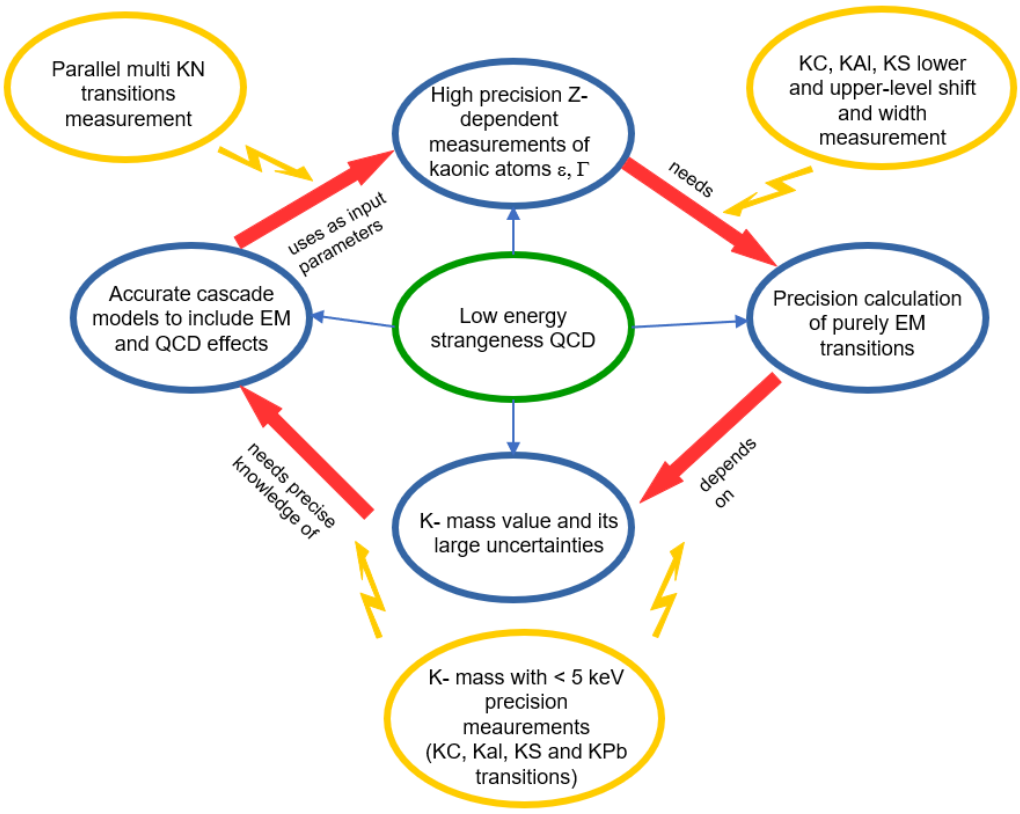


Fig. SA1. The impact of EXOTIC in low-energy strangeness QCD.

The kaonic atoms are exotic systems where a K- replaces an electron in a normal atom and, being formed in an excited state, cascades down towards lower levels emitting radiation. These lower levels are close enough to the nucleus to be shifted and broadened by the strong interaction; their pure QED energies can be calculated with extreme precisions, so measurements of the radiation provide information on strong interaction at the (nearly) threshold, serving as experimental inputs for the development of all the theoretical models describing the interaction of antikaons and nucleons.
Kaonic atoms were widely investigated in the 1970s and 1980s when a systematic study of kaonic atoms along the periodic table was performed [1]. This allowed for setting up a **dataset which served, until nowadays, as the main experimental input for many theoretical models**. Recently, new measurements performed at the DAFNE collider of INFN Laboratories of Frascati (LNF) and at the K1.8BR beamline at J-PARC revealed how **the older data cannot be trusted**; the recent re-measurements of kaonic atoms (KH and KHe) solved a series of puzzles, but also **cast doubts on the full set of existing data**, serving as basis of all models and theories of the strong interaction in systems with strangeness at low energies [2].
Furthermore, many of the earliest measurements of the strong interaction-induced shifts and widths of the lower levels in atoms like **carbon, aluminium and sulphur suffer from exceptionally large uncertainties** and, in some cases, they are **in contrast with the most recent theoretical models [3].**
The extraction of the strong interaction-induced shifts and widths of the kaonic atomic levels **relies on a chain of interrelated**

elements, the first one being a precise determination of the K- mass: the present published value (see Fig. SA2) results from an average of several measurements where the error has been artificially increased by a scale factor S=2.5 because of a large disagreement between the two most recent and precise results, both obtained from kaonic atoms spectroscopy [4].

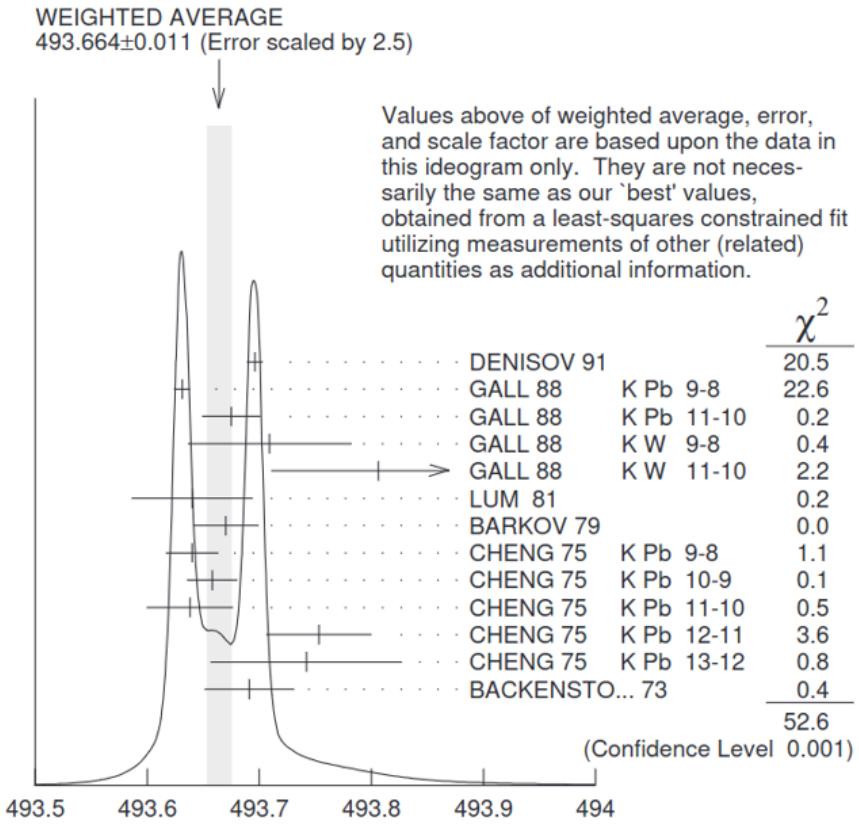


Fig. SA2: value of the weighted average K- mass published in the PDG (https://pdg.lbl.gov/2020/listings/rpp2020-list-K-plus-minus.pdf)

A more precise measurement of the charged kaon mass could be performed using several high-n transitions in intermediate and heavy mass kaonic atoms, where the strong interaction does not play a role, but accurate cascade models are required to properly include QED effects and to reduce the systematic errors. In particular, **KPb transitions are ideal for such purpose** and the KPb(9->8) one is exactly that used in one of the two critical measurements from which the present value is derived [4]. However, the development of such cascade models relies on high-precision kaonic atom measurements as inputs which, in turn, require the K- mass value as an input parameter for the calculations of their pure electromagnetic values resulting in a vicious cycle.

The situation, which is schematically depicted in Fig. SA1, is then in impasse and the way out is by breaking this vicious cycle by performing precise multiple crucial measurements.

The special behaviour of kaonic carbon was also stressed in [5], where the authors noticed that “In comparing between calculation and experiment we note that for almost all species of kaonic atoms in the database the absorptions from the upper state are of the order of 10–15% of all absorptions, as deduced from the measured upper level to lower level radiation yields. Carbon is an exception with, by far, the lowest radiation yield (of 0.07 ±0.013). Atomic cascade calculations show that for C about 75% of the absorptions take place from the upper state. Therefore, the calculated single-nucleon absorption fractions should be close to the upper-level points for C and close to the lower-level points for all the other species”.

More recent theoretical works established that good fits to kaonic atom data over the entire periodic table required additionally sizable absorptive and dispersive phenomenological terms motivated by the interaction of kaons with two nucleons (KNN) [3].

The kaonic aluminum is also an open issue, since the small width of the KAl(4->3) transition (see Fig. SA3) is an indication of the fact that the K- absorption may also occur from level n=2, but the KAl(3->2) transition has never been measured until now, probably due to a very low yield.

Concerning kaonic sulfur, in a recent workshop on strangeness nuclear physics organized in 2021 [6], updated fits with new theoretical approaches to the existing kaonic atoms data were recently shown, evidencing **great problems when fitting**

KS data [7]. More in general, several contributions remarked how “It would be desirable to revise some kaonic atom data ” (J. Hrtankova [7]) and “It is high time for new experiments” (E. Friedman [8]).

It is then evident how **new and more precise measurements of kaonic atoms are mandatory**. In particular, data from **kaonic carbon, aluminium and sulphur are highly demanded** and it would be crucial to have simultaneous measurements of different transitions, which will be also strategic to tune the QED cascade models used to determine the purely EM energies of the kaonic atoms levels.

Table 1
Compilation of K^- atomic data

Nucleus	Transition	ϵ (keV)	Γ (keV)	Y	Γ_u (eV)
He	3→2	−0.04 ±0.03	–	–	–
		−0.035±0.012	0.03 ±0.03	–	–
Li	3→2	0.002±0.026	0.055±0.029	0.95±0.30	–
Be	3→2	−0.079±0.021	0.172±0.58	0.25±0.09	0.04±0.02
^{10}B	3→2	−0.208±0.035	0.810±0.100	–	–
^{11}B	3→2	−0.167±0.035	0.700±0.080	–	–
C	3→2	−0.590±0.080	1.730±0.150	0.07±0.013	0.99±0.20
O	4→3	−0.025±0.018	0.017±0.014	–	–
Mg	4→3	−0.027±0.015	0.214±0.015	0.78±0.06	0.08±0.03
Al	4→3	−0.130±0.050	0.490±0.160	–	–
		−0.076±0.014	0.442±0.022	0.55±0.03	0.30±0.04
Si	4→3	−0.240±0.050	0.810±0.120	–	–
		−0.130±0.015	0.800±0.033	0.49±0.03	0.53±0.06
P	4→3	−0.330±0.08	1.440±0.120	0.26±0.03	1.89±0.30
S	4→3	−0.550±0.06	2.330±0.200	0.22±0.02	3.10±0.36
		−0.43 ±0.12	2.310±0.170	–	–
		−0.462±0.054	1.96 ±0.17	0.23±0.03	2.9 ±0.5
Cl	4→3	−0.770±0.40	3.80 ±1.0	0.16±0.04	5.8 ±1.7
		−0.94 ±0.40	3.92 ±0.99	–	–
		−1.08 ±0.22	2.79 ±0.25	–	–
Co	5→4	−0.099±0.106	0.64 ±0.25	–	–
Ni	5→4	−0.180±0.070	0.59 ±0.21	0.30±0.08	5.9 ±2.3
		−0.246±0.052	1.23 ±0.14	–	–
Cu	5→4	−0.240±0.220	1.650±0.72	0.29±0.11	7.0 ±3.8
		−0.377±0.048	1.35 ±0.17	0.36±0.05	5.1 ±1.1
Ag	6→5	−0.18 ±0.12	1.54 ±0.58	0.51±0.16	7.3 ±4.7
Cd	6→5	−0.40 ±0.10	2.01 ±0.44	0.57±0.11	6.2 ±2.8
In	6→5	−0.53 ±0.15	2.38 ±0.57	0.44±0.08	11.4 ±3.7
Sn	6→5	−0.41 ±0.18	3.18 ±0.64	0.39±0.07	15.1 ±4.4
Ho	7→6	−0.30 ±0.13	2.14 ±0.31	–	–
Yb	7→6	−0.12 ±0.10	2.39 ±0.30	–	–
Ta	7→6	−0.27 ±0.50	3.76 ±1.15	–	–
Pb	8→7	–	0.37 ±0.15	0.79±0.08	4.1 ±2.0
		−0.020±0.012	–	–	–
U	8→7	−0.26 ±0.4	1.50 ±0.75	0.35±0.12	45 ±24

Fig. SA3: Summary of the existing measurements of lower transitions in kaonic atoms

The energies and the widths of the kaonic atoms transitions of interest span over a very wide range, going from a few keV up to almost 1 MeV, depending on the Z of the selected atom and on the two levels between which the transition occurs. The choice of the proper detector to measure them is then crucial, and it must be taken as a function of their efficiency, energy resolution, linearity, and a well-known calibration function, as well as the capability to work in typically high background particle accelerators.

In this context, CZT detectors represent ideal candidates and have recently experienced a technological boost that promoted them as serious competitors to the more established Silicon (Si) and High-Purity Germanium devices (HPGe) [9].

The advantage with respect to Si is the much wider energy range since the typical 450-500 mm thick Si detectors are limited to 20 keV; very recently, attempts to use 1 mm thick detectors have been ongoing but, even if used in a stacked configuration reaching an overall thickness of 2 mm, reasonable efficiencies can be only obtained up to 50-60 keV. Another advantage is the capability of CZT detectors to work at almost room temperature; for more stability and a slightly improved energy resolution they can be cooled down to lower temperatures (about 5°) but they do not require cryogenic ones. Moreover, the use of unipolar geometries (hemispherical, co-planar geometries) allows electron sensing properties, a key element to reduce the hole influence in the signals and the rise time of the pulses.

Beyond the obvious advantage of the temperature working point, the great benefits with respect to Germanium are represented by a much better time resolution and a much stronger radiation hardness. Typical time resolution of HPGe is of the order of ms, with respect to that of CZT (presently) at the μs level for hemispherical.

Within the EXOTIC project, with a new geometry, front-end and digital readout electronics, **we aim to dramatically improve this time resolution pushing it below 1 ns**; this would be crucial to enhance their triggering and background reduction capabilities, ensuring the success of the proposed kaonic atoms' measurements in accelerators environment.

In the last years, several detector geometries have been studied to overcome the poor hole transport properties in CZT, which could lead to inefficient and low-performance devices.

Among all the electrode geometries, the quasi-hemispherical configuration allows the collection of charges over large detector volumes with a single readout channel. Their fabrication procedure is not critical since all electrodes can be deposited in a single step after photolithography.

Currently, the best resolution achieved with the quasi-hemispherical detectors is 4.4% and 2.9% at 59.5 keV and 122 keV respectively.

The coplanar grid sensing technique has shown a considerable enhancement in the spectral performance of large-volume CZT detectors removing the limitations due to poor hole collection and providing an adequate correction for electron trapping. This technique, combined with recent advances in CZT manufacturing, yields large-volume high-resolution room-temperature gamma-ray sensors for a wide range of applications such as nuclear material monitoring, radioisotope identification, gamma-ray astronomy, and medical diagnostics. The coplanar grid technique can currently provide an energy resolution of less than 1% FWHM at 662 keV for a cubic cm device.

The IMEM-CNR, a partner of the University of Palermo (which will be responsible for the CZT units production), has developed a novel simulation system [10], combining an MC simulator, a Finite Elements Method (FEM) calculator and a numerical computation software, capable of reproducing the radiation-semiconductor interaction, the weighting/electric fields, and the carrier transport/signal induction, respectively. It allows the prediction of the best design detector in terms of dimension, contact geometry and operating conditions, without requiring calibration measurements, providing large benefits in terms of time consumption and development costs.

The system simulates the transport of charges generated by photon-matter interactions drifting inside the electric field of the biased CZT sensor. The charge motion induces a signal on the collection electrode, and the software calculates trajectories of both carrier types by means of ODE (Ordinary Differential Equation) functions. Once motion equations are known, the software calculates the induced current for both electrons and holes according to the Ramo-Shockley theorem. The total charge is calculated by integrating simulated current signals. The last step is the binning of the total charge values and the extrapolation of the spectrum. The best detector geometry will be determined by comparing spectra generated by several simulated devices. The power of this technique is that many possible sensor geometries can be considered.

The information provided by this tool, such as electric field, weighting field, and induced current transients, can be very useful to improve the readout electronics or digital corrective system based on event pulse shape. This tool has proven to be fast and stable on several occasions and nowadays it is performing simulations for detector optimization in the X-ray energy range.

It can be expected that the best detector for EXOTIC will be chosen between quasi-hemispherical and coplanar grid detectors, and that thanks to the optimization of geometric parameters, the use of ultra-low-noise readout electronics and the implementation of Pulse Shape and Height Analysis, the resolution will be reduced down to FWHM < 1.2% and < 0.8% at 63 keV and 162 keV, respectively.

State of art: kaonic atoms with CdZnTe detectors

CZT detectors have never been used, so far, to measure radiative transitions of kaonic atoms. Their behaviour in high electromagnetic background environments is not well known and ad-hoc crystals and detectors must be realized to maintain a high efficiency in the whole energy range, without worsening the spectroscopic resolution.

Also, **it is crucial to define a perfect timing window** for the data acquisition to reject the typical huge background of particle accelerators. The SIDDHARTA-2 experiment already performed, in 2022 and 2023, preliminary tests in this direction with 1 cm2 hemispherical CZT detectors triggered with the Luminosity Monitor (LM) [11] of the experiment, able to distinguish events due to Kaons or Minimum Ionizing Particles (MIPs). **Very promising results were obtained**, like linearity better than 0.2% up to 511 keV (see Fig. SA4) and clear evidence of a time correlation between CZT and LM (see Figs. SA4 and SA5) [12] [13] [14].

It was also evidenced how, with the present hemispherical CZT technology, a timing window of 1 μs is needed to select the signal events; **this represents, nowadays, a strong limiting factor for the realization of measurements of low-yield transitions.** During these tests, in fact, the machine background suppression capabilities were investigated using a triple timing coincidence between the trigger system, based on plastic scintillators for TOF measurements, and the CZT detector. This investigation revealed how **pushing the time window from 1 μs to a few tens of ns** would translate in the possibility of **improving the rejection factors by an order of magnitude** or more, reaching the 10^7-8 level and **representing a**

breakthrough for high precision kaonic atoms measurements [13]. This will be possible thanks to the new 1 ns time resolution of the CZT, presently dominating the overall time window.

Given the present state of the art, both from the point of view of kaonic atoms and low energy strangeness QCD physics and of the CZT technology, the overarching objective of EXOTIC is to conduct an intertwined technological and scientific research in the field of the low-energy strong interaction with strangeness, by performing **unique measurements** of X-ray transitions in selected kaonic atoms employing **cutting edge CZT X-ray detectors** with high detection efficiency (above 70%), a <1% energy resolution, ~ 1 ns time resolution and a 10-4 linearity in the 25-350 keV range.

Scientific Objectives

In particular, the EXOTIC experiment aims to measure:

- The charged K- mass with a precision below 5 keV;
- The energy and width, for the first time, of the KAl(3->2) transition;
- The energies (E) and widths (W) of the KC(3->2), KS(4->3) and KAl(4->3) transitions with precisions better than 30 and 60 eV for E and W, respectively;
- The first measurements ever of the absolute yields of the Δn=1,2 transitions towards n=2,3 levels in KC, n=3,4,5 in KS and Kal and n=8,9,10,11 in KPb.

The present error on the published value of the K- mass is 13 keV [4], while the KAl(3->2) transition has **never been measured until now**. The KC(3->2) transition energy is measured with a precision of 80 eV, while there exist three and two measurements of KS(4->3) and KAl(4->3), respectively, not compatible among themselves and with large errors on the widths [1]. Most of the Δn=1,2 transitions in KC, Kal, KS, and KPb have **never been measured** and will provide new experimental inputs both to the Kaon-Nucleon(s) interaction theoretical models and to the EM atomic cascade ones [15] [16] [17].

Technological Objectives

EXOTIC plans to develop coplanar grid CZT detectors **with important advances with respect to the currently available knowledge and state-of-the-art.** The goal is to achieve the following performances in the whole 25-350 keV energy range:

- Ultra-fast time resolution (< 1 ns).
- High energy resolution at room temperature (≤ 1.2 % FWHM at 60 keV; ≤ 1% FWHM at 122 keV; ≤ 1% FWHM at 350 keV).
- High detection efficiency (100% at 63 keV, 70% at 162 keV, > 50 % at 300 keV).
- Modular geometry with individual detector-electronics units to be adapted for both fixed target and collision experiments.

The accomplishment of these performances is mandatory for the success of the EXOTIC experimental goals and represents a **technological challenge beyond the present state of the art**.

I strongly believe that, given the new technological opportunities and the unique features of both DAFNE and K1.8BR, **now is the perfect time** to finally address the increasing request for new kaonic atoms data and to overcome the present impasse in the low energy strangeness QCD.

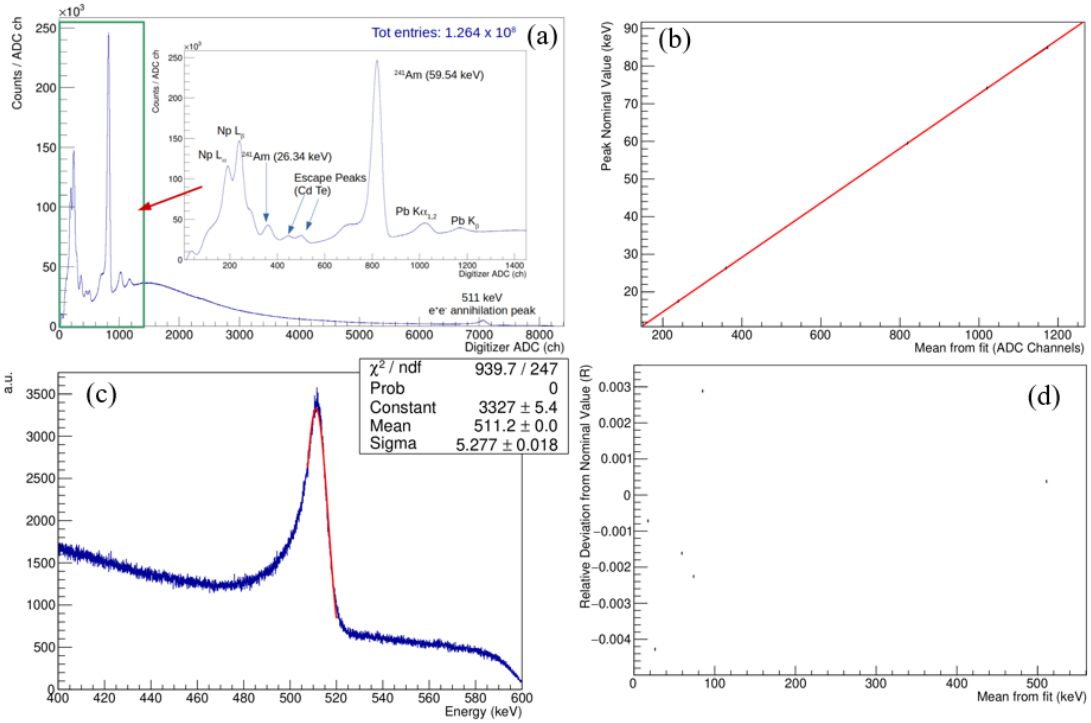


Fig. SA4: Results of preliminary tests with CZT detectors in DAΦNE. Overall spectrum (a), Calibration line (b), the resulting 511 keV peak (c) and the Relative deviation from Nominal Value after calibration (d). The calibration is performed using only peaks up to Pb Kβ, and a very good linearity is found when fitting the 511 keV peak on the calibrated spectra, also visible from the Relative Deviations from the Nominal Values of the peaks.

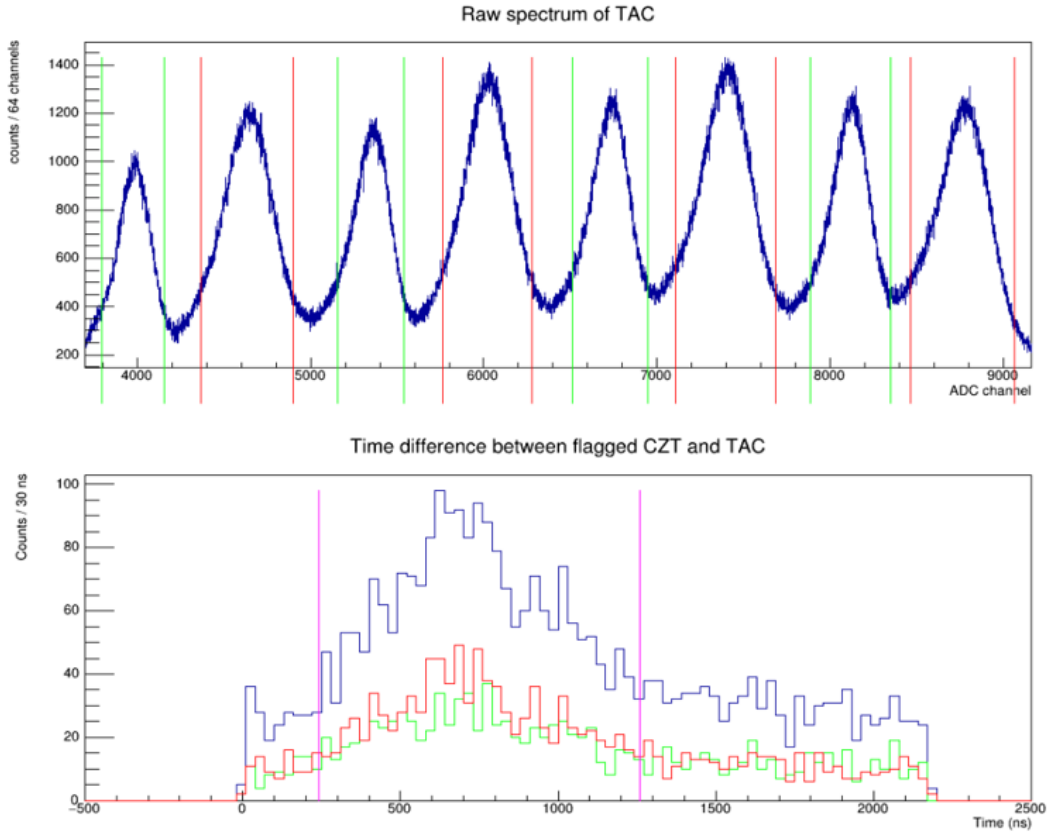


Fig. SA5: Top: SIDDHARTA-2 LM TOF spectrum, where windows for Kaons (green) and Minimum Ionizing Particles (MIPs, red) time selection are highlighted. Bottom: Timing spectrum of the CdZnTe with respect to the SIDDHARTA-2 LM (blue) with further selection of Kaons (green) and MIPs (red)

[1] doi={10.1016/0375-9474(94)90921-0}

[2] doi={10.1103/RevModPhys.91.025006}

[3] doi={10.1103/PhysRevC.106.065201}

[4] doi={10.1103/PhysRevD.98.030001}

[5] doi={10.1016/j.nuclphysa.2016.12.009}

[6] https://agenda.infn.it/event/25725/

[7] https://agenda.infn.it/event/25725/contributions/131349/attachments/79624/103640/foto-Jarka.pdf

[8] https://agenda.infn.it/event/25725/contributions/131346/attachments/79609/103625/Eli-Friedman-talk.pdf.

[9] doi={https://doi.org/10.3390/s90503491}

[10] doi={10.1016/j.nima.2020.163663}

[11] doi={10.1088/1748-0221/15/10/P10010}

[12] doi={10.48550/arXiv.2310.15062}

[13] doi={10.1140/epjs/s11734-023-00881-x}

[14] doi={10.3390/s23177328}

[15] doi={10.1103/physrevc.53.79}

[16] doi={10.1140/epjd/e2002-00208-x}

[17] doi={10.1103/PhysRevC.55.73}

Research Methodology:

The EXOTIC setup

The EXOTIC experimental apparatus has a modular concept based on single units, defined as a 1 cm2 hemispherical/coplanar-grid CZT detector coupled to its preamplifier board. This is **the key element** of the EXOTIC project because it allows the use of this **innovative X-ray detection technology** for exotic atom measurements both at colliders and in extracted beam experiments, like DAFNE and J-PARC for the case of kaonic atoms. EXOTIC can **easily switch between the two modalities** by using different structures to adapt the setup to the available spaces in the various facilities, through which the temperature stabilization at around 15-20°C could be achieved by means of a water-cooling system. The basic concepts of the two proposed setups are shown in Fig. M1.

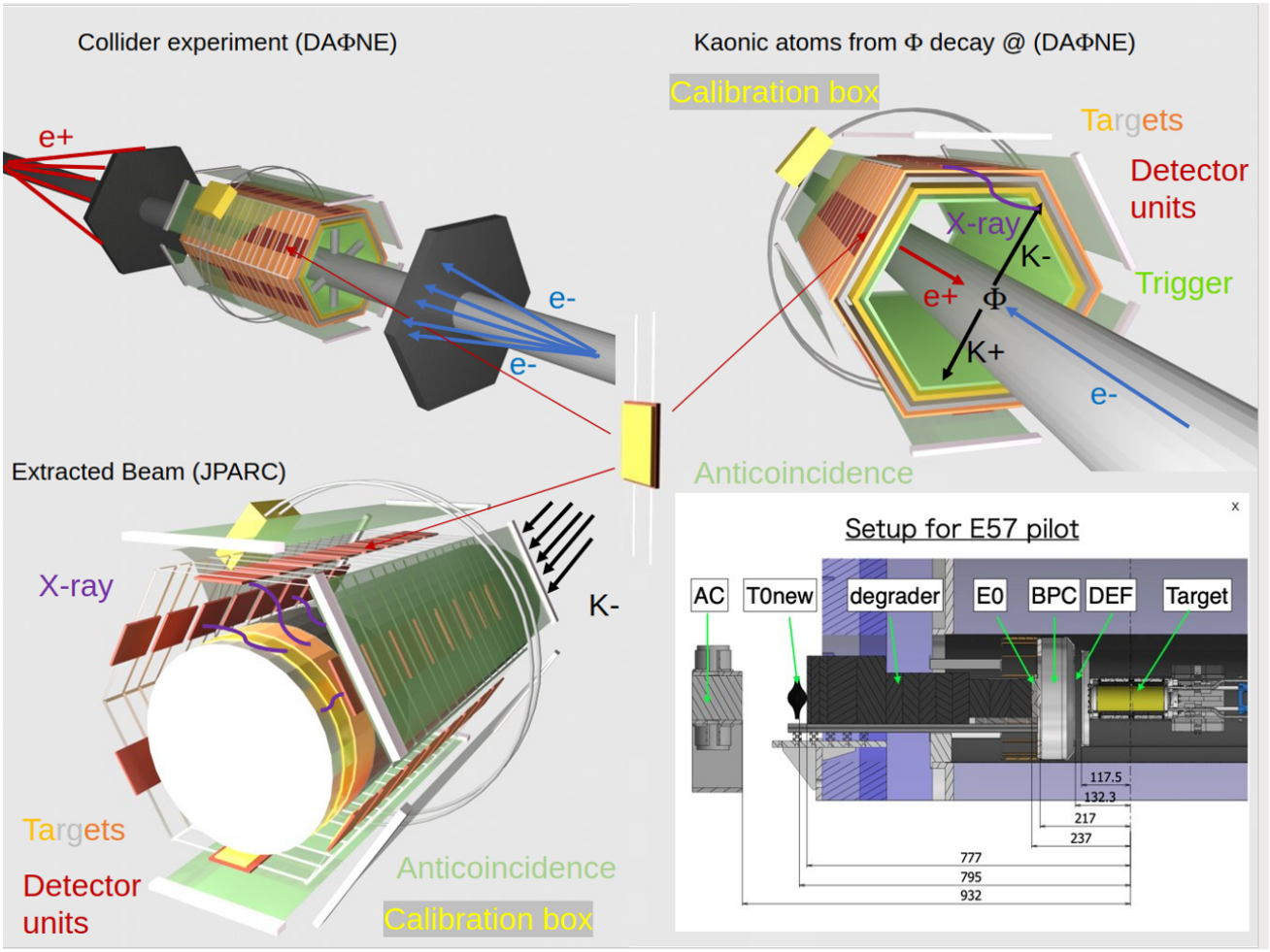


Fig. M1: (not in scale) sketch of the possible EXOTIC apparatuses to be used at DAΦNE (up) and JPARC (down). In the centre, the single unit is shown. More details in the text.

At the Interaction Region (IR) of DAFNE, the Φ mesons produced by the e+e- collisions decay into back-to-back low momentum (PK~ 100 MeV/c) K+K- pairs, with a mean rate of ~ 500 K/s. These pairs will be first detected by a layer of scintillators working as a trigger for the EXOTIC experiment and able to discriminate, by Time-Of-Flight differences between kaons and Minimum Ionizing Particles (MIPs), similarly to the tests performed in [1] [2] [3]. K- will then stop in the C, Al, S, and Pb targets arranged in a decreasing order of density to avoid self-absorption of the X-rays emitted by kaonic atoms. These latter will be then detected by the CZT units while a further external scintillator layer will serve as anticoincidence to detect the MIPs passing through the detectors and producing fake signals. A double Pb shielding structure will be placed in front of the two ends of the IR to suppress the electromagnetic showers produced in the quadrupoles present in those regions.

At J-PARC, the E57 experiment [4] will measure in 2027-2028 X-rays from liquid kaonic deuterium, exploiting the ~ 10^8/s K- extracted from the main proton beam. Almost 90% of these K- will stop in the carbon degrader of the experiment which, for EXOTIC, will be modified with the insertion of a series of targets, with thicknesses chosen to have an equivalent density to the original degrader's one to not interfere with the E57 experiment. The CZT detectors will surround these targets and the degrader to detect the X-rays emitted from the kaonic atoms formed inside. An external scintillator layer will be used as anticoincidence while a series of existing detectors (A0, T0new, E0 and BPC) will ensure the trigger and the beam monitoring [4].

In both cases, a **crucial role** will be played by the ~1 ns time resolution of the CZT detectors, for a highly efficient background suppression.

The CZT units will be placed on six hexagonal prism structures, each face having 50 detectors for a total active area of 300 cm2, the dimensions of which are optimized to match the available space.

The long-term stability of the CZT, **never tested up to now in accelerator environments for data-taking periods of more than several days**, will be monitored using **periodical calibrations** performed with radioactive sources, like 241Am, with emission lines in the range of interest. The sources will be enclosed in a remotely on-off switchable shielding box and placed on a motorized circular rail around the hexagonal prism to shine all six faces.

To validate the EXOTIC feasibility, a 15,6 cm2 CZT detector has been inserted in the GEANT4 MC simulation code of SIDDHARTA-2, the experiment presently installed at DAFNE, to estimate the expected results **in real beam conditions**. The schematic view is shown in Fig. M2. MC simulations corresponding to 10 pb^-1 of integrated luminosity (~2 days of data taking) were run, as a test, with a C/Al and a Pb/S mixed targets attached to one of the two scintillators of the SIDDHARTA-2

Luminometer (LM) [5], while the detectors were placed at a distance of 216 mm from the IR surrounded by Pb bricks for shielding. Targets' thicknesses were optimized to stop in each one a similar fraction of K-.

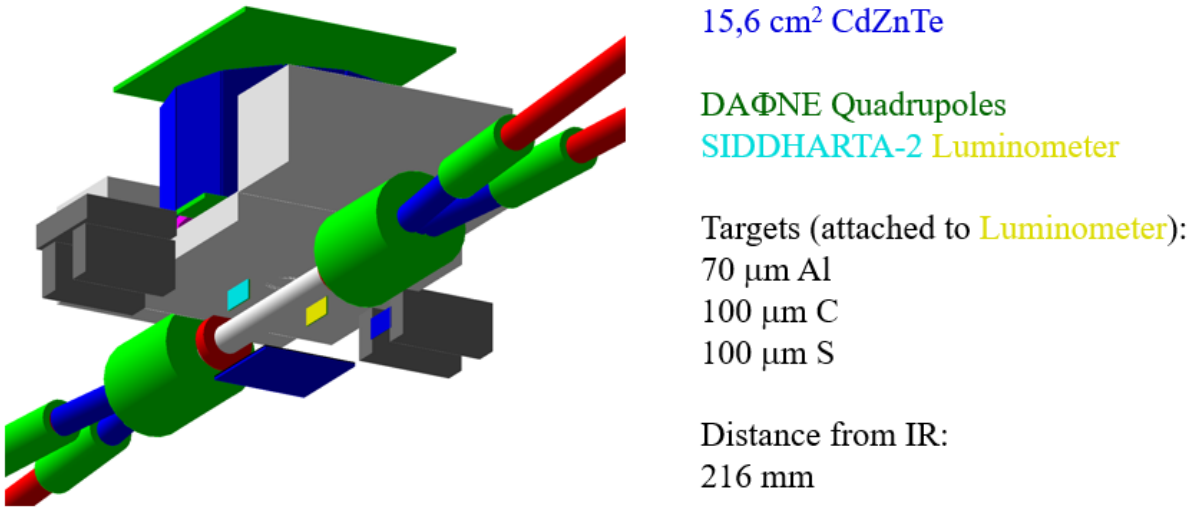


Fig. M2: Details of the GEANT4 simulations of 15,6 cm2 CdZnTe inserted in the main SIDDHARTA-2 simulation

In the MC code, Φ mesons were generated with true momentum distribution and all the kaonic atoms X-rays of interest were isotropically generated for each K- stopping in one of the targets. The electromagnetic background was not present, but a proper estimate of the signal-to-background ratio has been made based on the several data acquired by the SIDDHARTA-2 collaboration at DAFNE.

The main results are shown in Fig. M3 where, as an example, the spectrum obtained for the C/Al target after the selection of K+K- coincidences in the two scintillators of the LM is shown.

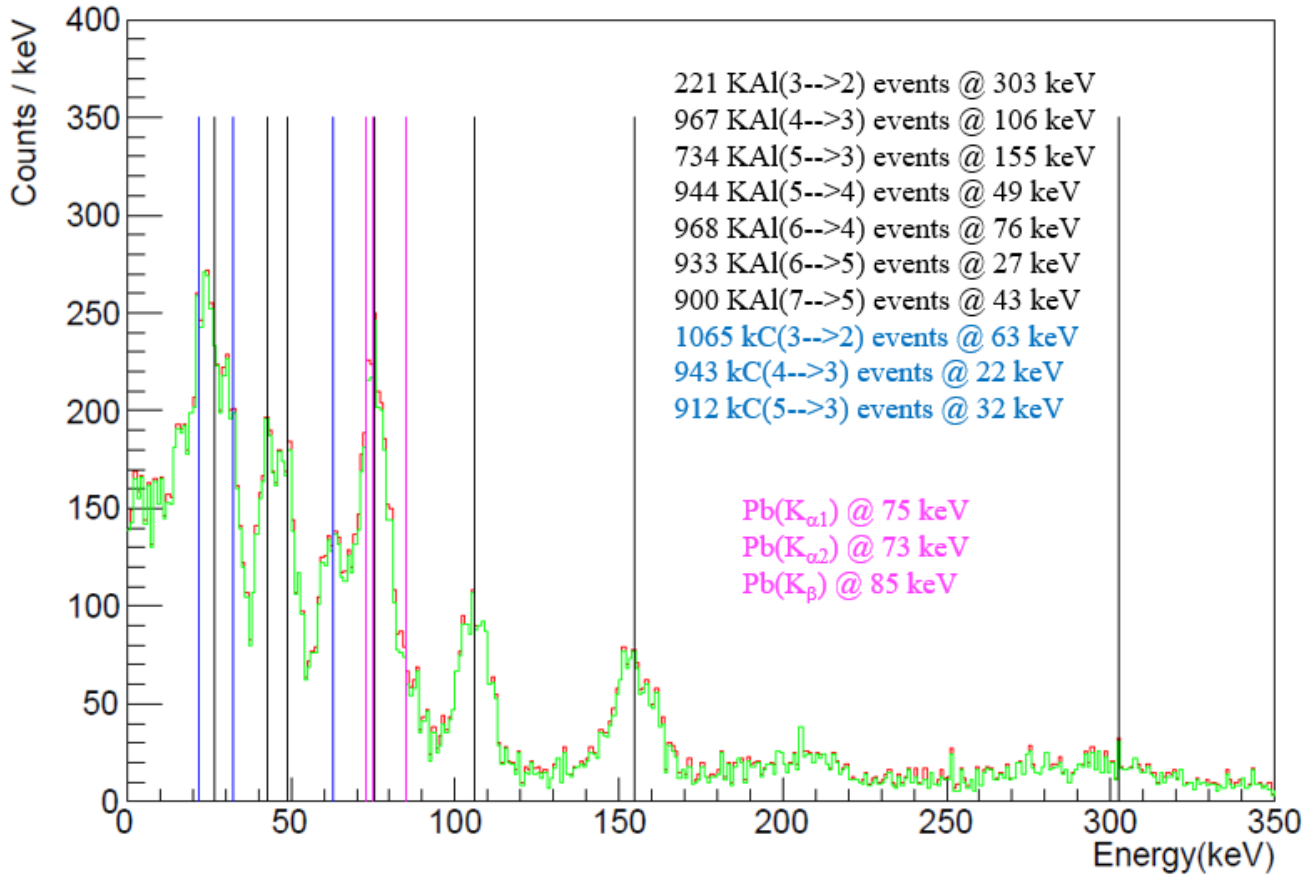


Fig. M3: GEANT4 MC simulation of a 15,6 cm^2 CZT detector within the SIDDHARTA-2 experiment, placed at 216 mm from the IR. The statistics corresponds to an integrated luminosity of 10 pb^{-1} (2 days of data taking). The kaonic lines are generated in the target assuming a 100% yield per K- stop, while the pink lines indicate X-ray fluorescence peaks from Pb shielding activated by particles lost.

From the number of signal events obtained from the MC simulations, the errors in the transitions' energies and widths can be estimated. from basic statistical considerations and are reported in Fig. M4.

The expected number of events in 60 days assumes a very conservative integrated luminosity of 5 pb-1 /day. The yields for the lower transitions are taken from the kaonic atoms' database [6], while for the upper ones, no experimental data are available. For the ΔN=1 (green) upper-level transitions the yields are set, conservatively, equal to the lowest ones even if higher n transitions usually have higher yields. In case of ΔN=2 (red), a further factor of 2 less is assumed. Another assumption, based on experimental observations, is that the natural line widths of the upper levels are orders of magnitude smaller than those of the lower levels and, in our case, the final resolution is totally dominated by the detector resolution. Finally, the errors are estimated from the peak resolution and the number of events assuming the highly conservative scenario of a signal-to-background ratio of 1/5, as well as for a 'worst case' scenario of a signal-to-background ratio of 1/50.

Transition	Energy (keV)	events / 10 pb^{-1} with 15,6 cm^2	Yield / K^- stop	Expect events in 60 days with 300 cm^2	Natural Linewidth Γ (keV)	CdZnTe resolution σ (keV)	Convoluted resolution σ (keV)	estimated error on the energy (keV) for S/B=1/5	estimated error on the width Γ (keV) for S/B=1/5	estimated error on the energy (keV) for S/B=1/50	estimated error on the width Γ (keV) for S/B=1/50
KC(3-->2)	63	1065	0,07	28386	1,8	1,26	1,475	0,020	0,039	0,062	0,124
KAl(4-->3)	106	967	0,55	202512	0,45	2,12	2,129	0,011	0,021	0,033	0,067
KS(4-->3)	161	940	0,22	78743	2,2	3,22	3,353	0,027	0,053	0,084	0,169
KC(4-->3)	22	943	0,07	25135	0	0,44	0,440	0,006	0,012	0,020	0,039
KC(5-->3)	32	912	0,035	12154	0	0,64	0,640	0,013	0,026	0,041	0,082
KAl(5-->4)	49	944	0,55	197695	0	0,98	0,980	0,005	0,010	0,016	0,031
KAl(6-->4)	76	968	0,27	99518	0	1,52	1,520	0,011	0,022	0,034	0,068
KAl(6-->5)	27	933	0,55	195392	0	0,54	0,540	0,003	0,005	0,009	0,017
KAl(7-->5)	43	900	0,27	92527	0	0,86	0,860	0,006	0,013	0,020	0,040
KS(5-->4)	74	1590	0,22	133193	0	1,48	1,480	0,009	0,018	0,029	0,057
KS(5-->3)	235	560	0,11	23455	0	4,7	4,700	0,069	0,137	0,217	0,434
KS(6-->5)	40	1350	0,22	113088	0	0,8	0,800	0,005	0,011	0,017	0,034
KS(6-->4)	115	1310	0,11	54869	0	2,3	2,300	0,022	0,044	0,069	0,139
KS(7-->5)	65	1700	0,11	71204	0	1,3	1,300	0,011	0,022	0,034	0,069
KPb(9-->8)	289	170	0,79	51137	0	5,78	5,780	0,057	0,114	0,181	0,361
KPb(10-->9)	207	190	0,4	28938	0	4,14	4,140	0,054	0,109	0,172	0,344
KPb(11-->10)	153	370	0,79	111299	0	3,06	3,060	0,021	0,041	0,065	0,130
KPb(12-->10)	269	130	0,4	19800	0	5,38	5,380	0,085	0,171	0,270	0,541
KPb(11-->9)	359	80	0,79	24065	0	7,18	7,180	0,103	0,207	0,327	0,655
KPb(12-->11)	116	360	0,79	108291	0	2,32	2,320	0,016	0,032	0,050	0,100
KPb(13-->11)	207	270	0,4	41123	0	4,14	4,140	0,046	0,091	0,144	0,289

Fig. M4: Error estimation from the KC and KAl MC simulations. The expected number of events in 60 days assumes a very conservative integrated luminosity of 5 pb^{-1} /day. The yields for the lower transitions are taken from the kaonic atoms' database [6], while for the upper ones no experimental data are available. For the ΔN=1 (green) upper-level transitions the yields are taken, conservatively, equal to the lowest ones even if higher n transitions usually have higher yields. In case of ΔN=2 (red), a further factor 2 less is assumed. Another assumption, based on experimental observations, is that the natural linewidths of the upper levels are order of magnitude smaller than those of the lower levels and, in our case, the final resolution is totally dominated by the detector resolution. Finally, the errors are estimated from the peak resolution and the number of events assuming a highly conservative scenario of a signal-to-background ratio of 1/5, as well as for a 'worst case' scenario of a signal-to-background ratio of 1/50.

From these estimates, it is clear how the goals of the EXOTIC projects for the KC, KAl, KS and KPb measurements could be already fulfilled with a two-month data-taking at DAFNE.

The link between the various transition energies and the Kaon mass value will be obtained through a code developed by Prof. Paul Indelicato [8] [9], one of the most recognized experts on kaonic atom cascade calculations. In his work, the energies of the kaonic atom transitions can be determined by calculating the total energies of the initial and final states using the Klein-Gordon equation; for each transition, several calculations using different Kaon mass values as input allow for deriving linear functions relating the transition energy and the Kaon mass.

An example derived for the KPb(9->8) transition can be found in Fig. M5

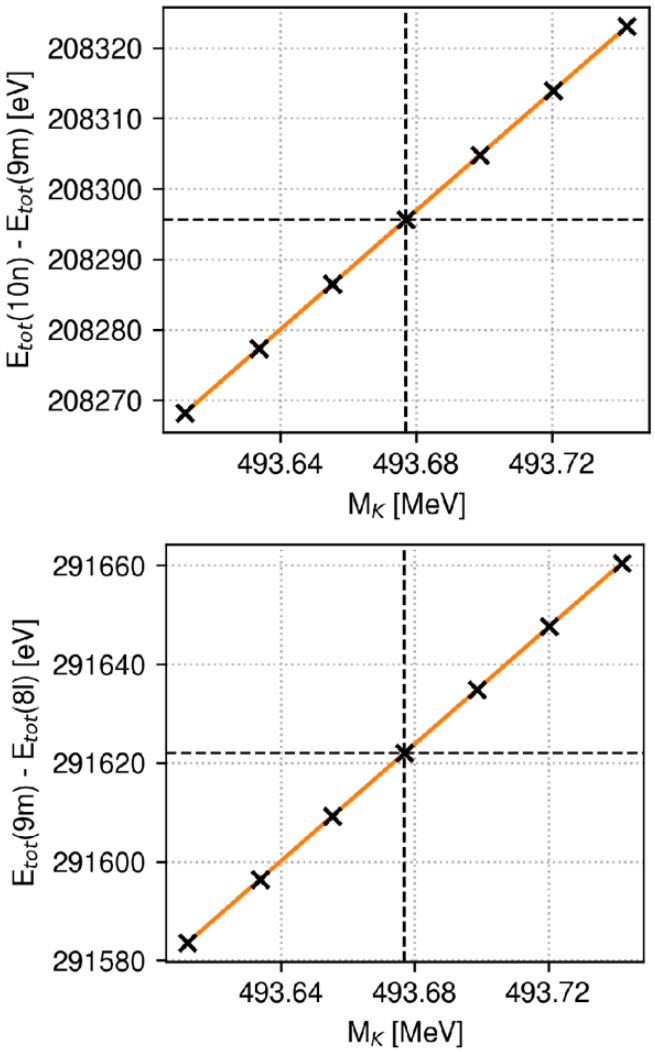


Fig. M5: KPb(10->9) and KPb(9->8) transition energies as a function of the Kaon mass obtained from calculations.

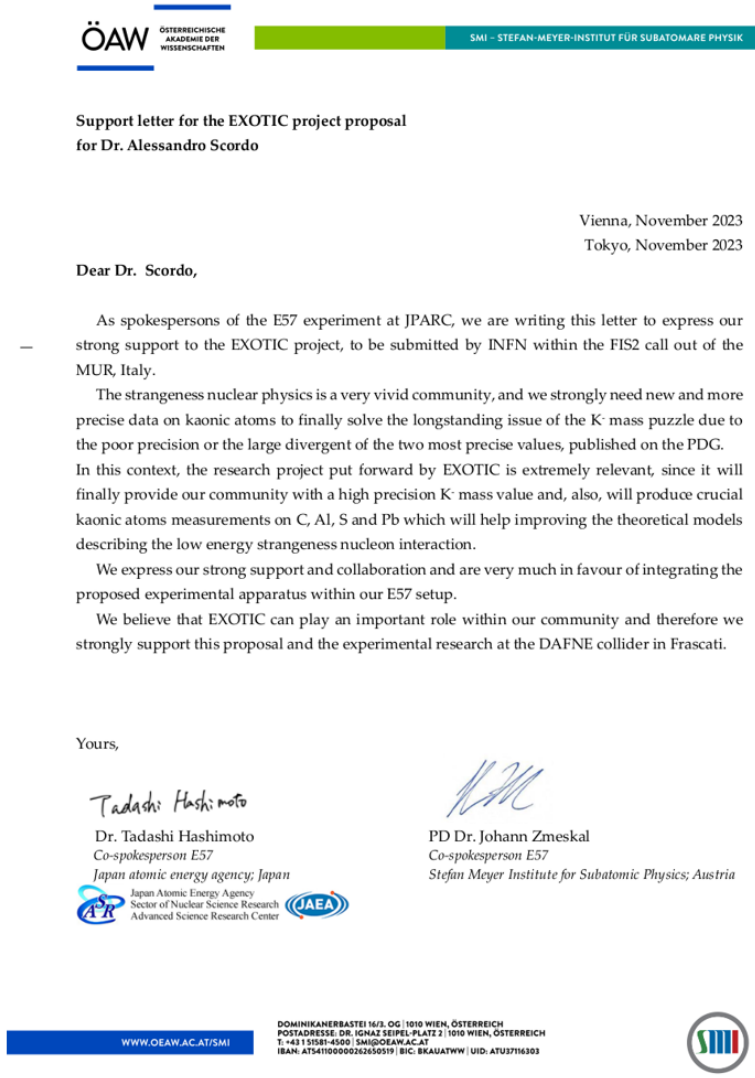
EXOTIC will then exploit these calculations to perform a parallel fit of all the measured transitions in which the various peak positions will be parametrized as a function of the Kaon mass, left as a free parameter. EXOTIC, **thanks to its wide energy range and a dedicated multitarget data taking**, will measure more than ten transitions towards upper levels in parallel, and such an approach will allow for a **huge reduction in statistical errors**.

To deliver the new Kaon mass with better than 5 keV precision, systematic errors have also to be kept under control. From the experience gained within the SIDDHARTA-2 collaboration, the systematic errors in such types of experiments are mostly due to the stability and linearity of the detectors.

Within the EXOTIC project, we plan to realize a **very low noise and temperature-stabilized** X-ray detection system, thanks to which **dramatically reducing the systematics**. Moreover, periodical calibrations will be performed with the system described before to reduce such systematic errors, **allowing for the success of the EXOTIC goal** of delivering a new measurement of the Kaon mass with less than 5 keV uncertainty.

Concerning the EXOTIC data taking at J-PARC, the 10^7-8 K-/s stopping in the target/degrader will ensure a statistic of orders of magnitude higher than that collectable at DAFNE. This will ensure the fulfillment of all the EXOTIC's goals, but if they will be already addressed during the DAFNE run, additional targets (O, Si, Co, Cu, etc.) may be used at J-PARC to improve the precisions, renewing and enlarge the present kaonic atoms' database.

The E57 spokespersons strongly support the EXOTIC proposals and they expressed their consensus on integrating its setup within the main E57 one (see Fig. M6)



ÖAW ÖSTERREICHISCHE AKADEMIE DER WISSENSCHAFTEN

SMI – STEFAN-MEYER-INSTITUT FÜR SUBATOMARE PHYSIK

Support letter for the EXOTIC project proposal
for Dr. Alessandro Scordo

Vienna, November 2023
Tokyo, November 2023

Dear Dr. Scordo,

As spokespersons of the E57 experiment at JPARC, we are writing this letter to express our strong support to the EXOTIC project, to be submitted by INFN within the FIS2 call out of the MUR, Italy.

The strangeness nuclear physics is a very vivid community, and we strongly need new and more precise data on kaonic atoms to finally solve the longstanding issue of the K^- mass puzzle due to the poor precision or the large divergent of the two most precise values, published on the PDG.
In this context, the research project put forward by EXOTIC is extremely relevant, since it will finally provide our community with a high precision K^- mass value and, also, will produce crucial kaonic atoms measurements on C, Al, S and Pb which will help improving the theoretical models describing the low energy strangeness nucleon interaction.

We express our strong support and collaboration and are very much in favour of integrating the proposed experimental apparatus within our E57 setup.

We believe that EXOTIC can play an important role within our community and therefore we strongly support this proposal and the experimental research at the DAFNE collider in Frascati.

Yours,

Tadashi Hashimoto

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Co-spokesperson E57
Japan atomic energy agency; Japan
Japan Atomic Energy Agency
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PD Dr. Johann Zmeskal
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Fig. M6: Support letter from the E57 spokespersons

The innovative EXOTIC X-ray detection system

A crucial point for the success of the EXOTIC experiment will be the background rejection both at J-PARC and DAFNE. This will be strongly improved thanks to the groundbreaking time resolution of the CdZnTe devices, coupled with the high efficiency and an energy resolution of a few per cent in the whole energy range of interest.

The detection system will be realized as follows:

-*Detectors*: Unipolar (i.e. electron-sensing: hemispherical; coplanar-grid geometries) CZT detectors will be developed, exploiting the advantages of CZT material, namely (i) high detection efficiency (70% for 5 mm thick at 162 keV), (ii) low Compton scattering (Compton probability of 10% at 120 keV) (iii) room temperature operation, (iv) short collection times (< 100 ns) and (v) good homogeneity and time-stability. CZT crystals grown by the Traveling Heater Method (THM), characterized by enhanced-charge transport properties of electrons (mobility-lifetime products μτ > 10^−2 cm2/V), will be used for the detector fabrication.

-*Front-End Electronics (FEE)*: The Front-End Electronics (preamplifiers) to read out the outputs of the detectors will be an ASIC with a low-noise Charge Preamplifier (CP). The ASIC design criteria will be focused on ideally matching the pixel characteristics (leakage current and capacitance) and reaching high speed, low noise and low power consumption. It will be designed in the well-established 0.35 μm technology.

-*Digital Pulse Processing (DPP) Electronics*: The output pulses from the ASIC will be processed by DPP electronics. As well documented in the literature, several detection systems are currently developed by using DPP techniques, where the detector-preamplifier output signals are directly fed into fast digitizers and real-time processed by using digital algorithms. As widely recognized, the digital approach gives more benefits than the analog one, among which: (i) the possibility to implement optimized filters and signal discrimination procedures, which are challenging or impossible to realize in an analog approach, (ii) stability and reproducibility (insensitivity to pick-up noise as soon as the signals are digitized), (iii) flexibility at different rate

conditions and (iv) the possibility to perform multi-parameter analysis for detector performance enhancements and new applications. Digital electronics foresees synchronization with anti-coincidence systems for background reduction.

The EXOTIC Workplan

The EXOTIC Workplan is organized in 6 Work Packages (WP), with 3-7 tasks each (T), summarized in the GANTT chart of Fig. M7; for each of them, the Main Responsible (MR) is indicated with a color-referenced filled cell, while the 5 Milestones (M) are indicated with blue bars at their month of delivery.

In Fig. M8, the PERT diagram of EXOTIC is shown, where the tasks of each member of the team are highlighted in color-referenced frames, while the whole diagram is embedded in the 'PI Supervision' frame.

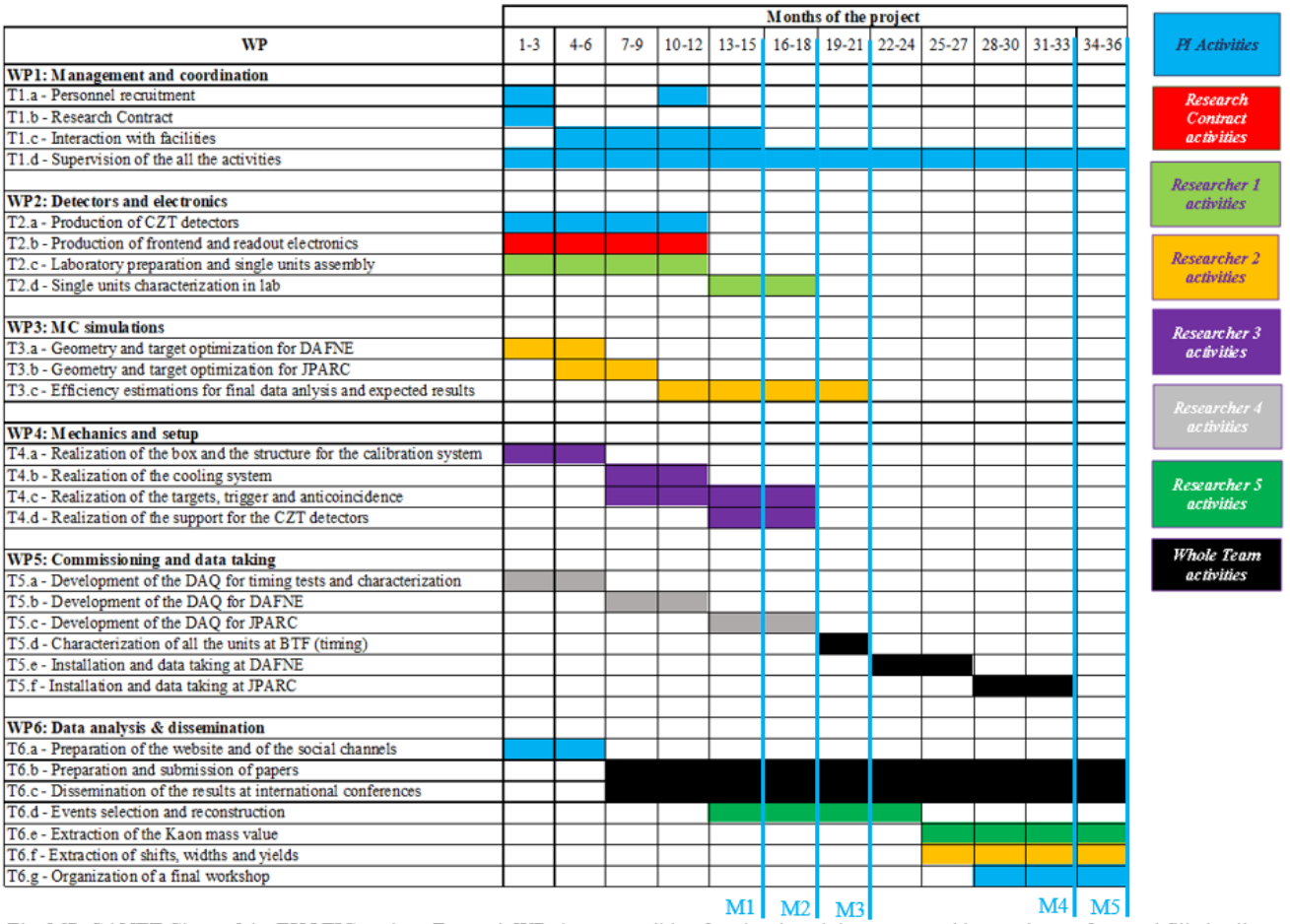


Fig. M7: GANTT Chart of the EXOTIC project. For each WP, the responsible of each sub-task is represented by a colour-referenced filled cell. In light blue, the 5 Milestone are reported at their month of delivery.

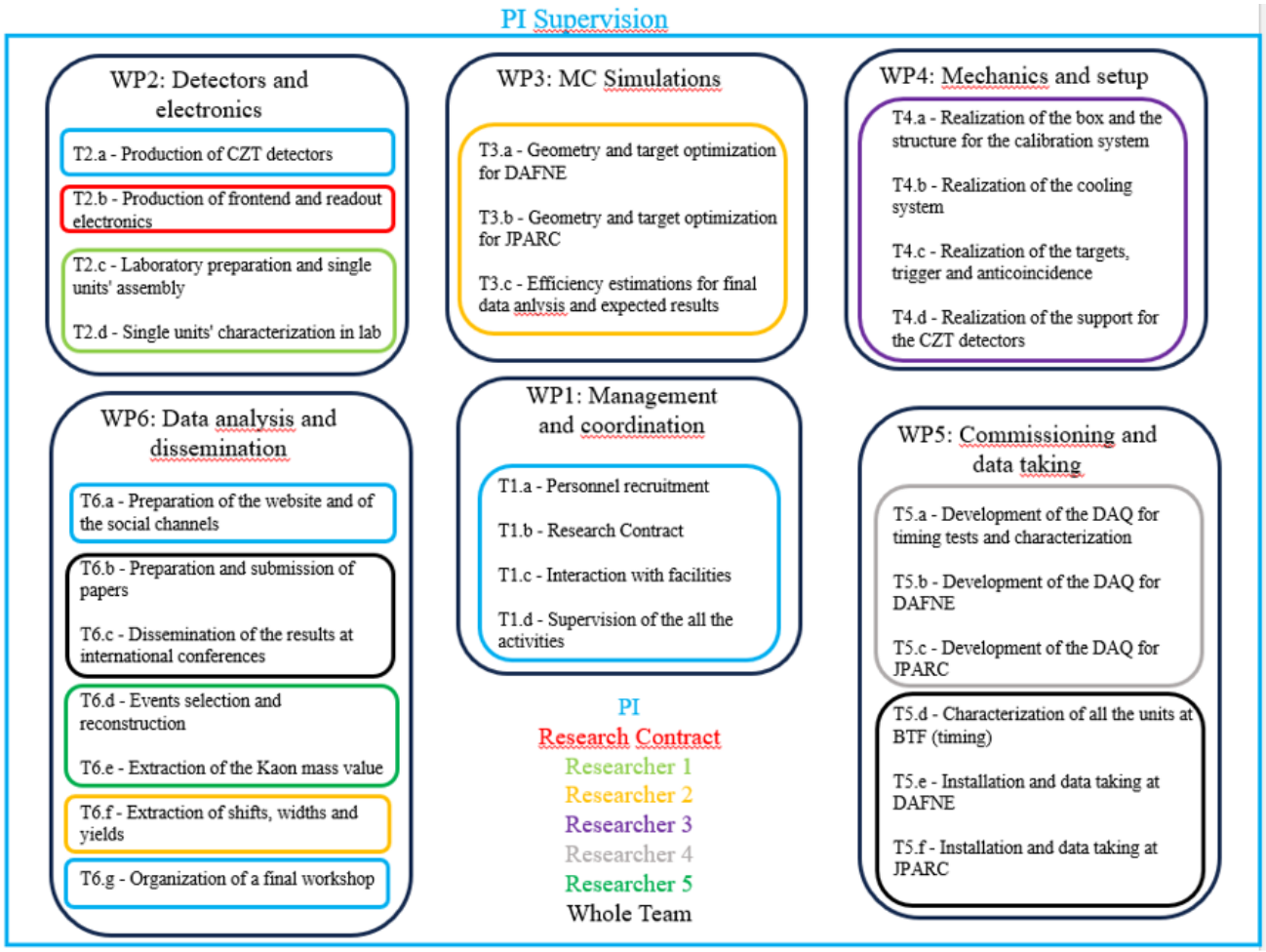


Fig. S9: PERT diagram of EXOTIC. The task of each member of the team are highlighted in color-referenced frames, while the whole diagram is embedded in the 'PI Supervision' frame.

[1] doi={10.1140/epjs/s11734-023-00881-x}

[2] doi={10.3390/s23177328}

[3] doi={10.48550/arXiv.2310.15062}

[4] https://indico.ectstar.eu/event/182/contributions/4281/

[5] doi={10.1088/1748-0221/15/10/P10010}

[6] doi={10.1016/0375-9474(94)90921-0}

[7] https://indico.ectstar.eu/event/182/

[8] doi={10.1088/0022-3700/20/4/007}

[9] doi={10.1103/PhysRevA.71.032501}

[10] doi={10.1103/RevModPhys.91.025006}

[11] doi={10.1016/j.nuclphysa.2016.12.009}

[12] doi={10.1103/PhysRevC.106.065201}

PROJECT SCHEDULE

Milestone	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Total Cost
M1	Definition of the final data-taking schedule at DAFNE and JPARC	15	3	6	3	0	0	0	
M2	Characterization of all the single units in laboratory	18	6	6	6	0	0	0	1380130
M3	Characterization of the whole system at the BTF facility of LNF	21	0	0	0	3	0	0	174405
M4	End of the data taking activity	33	0	0	0	3	6	3	779345
M5	Delivery of the scientific results of EXOTIC	36	0	0	0	0	6	6	43540

GANT CHART

GANTT_EXOTIC.pdf

EXPECTED IMPACT

The anticipated impact of the EXOTIC project is substantial, encompassing a broad spectrum of scientific and technological domains due to its multidisciplinary nature.

The scientific impact of EXOTIC:

- From the kaonic atoms X-ray transitions a new determination of the charged Kaon mass, which is at present known with a 3x10^-5 precision [1], with a precision better than 5 keV will be obtained also solving the puzzling divergence between the existent measurements. This new and more precise value will have **a dramatic impact on the low energy strangeness QCD** since the electromagnetic energies of kaonic atoms levels are sensitive to this parameter. It will also have **additional physics relevance**, being directly connected to fundamental parameters like CKM matrix elements, tests of CP and CPT with charged kaons, kaonic beta decay and form factors, and the physics of D mesons [2] [3] [4].

-The measurements of KC(3->2), KAl(4->3) and KS(4->3) will provide **new and more stringent constraints for all the models** describing the interaction between kaons and nucleons, with a better understanding of the role of kaon-multinucleon interactions **[5].**

-The EXOTIC data will represent **a gold mine** for all the theoretical models describing the atomic cascades, which are built and **rely on the experimental inputs provided by kaonic atom measurements [6] [7] [8] [9].** These calculations are in turn the basis of all exotic atoms research programs, and this will have EXOTIC impact **not only in kaonic atoms physics**. The key role of exotic systems in understanding fundamental mechanisms and the necessity of a new interplay among the various exotic atoms community is also evident from the recently approved KAMPAI (Kaonic, Antiprotonic, Muonic, Pionic and “onia” exotic Atoms: Interchanging knowledge and recent results) workshop at ECT* for 2024.

-The new data on kaonic atoms will contribute to the hot topic research on **the neutron stars EOS and the possible related presence of strangeness**, as proved by the recent ROCKSTAR workshop at ECT* [10].

As an overall result, the kaonic atoms measurements performed by EXOTIC will break the present impasse in the low energy strangeness QCD, connected to a lack of data, which prevents from a proper tuning of the theoretical models.

The technological and societal impact of EXOTIC:

In addition to the physics and technological purposes of the project, its success will trigger large benefits in several application fields:

-*bulk material analysis with muonic atoms*: the developed system can be very appealing for non-destructive bulk material analysis using muonic hard X-rays.

-*civil security:* nowadays, X-ray scanners only allow for very approximate discrimination between light elements and heavy ones and are relatively blind to drugs or explosives; on the contrary, the new ERPC systems will allow fine material discrimination so that it will be possible to discriminate among different liquids or increase the sensitivity of light elements (drugs or explosives).

-*food inspections:* the presence of foreign bodies inside processed food represents a huge issue for food companies, and the available scanners cannot reveal tiny or light contaminants, or become useless in the case of not-homogeneous medium -*material recycling:* the novel systems will be able to differentiate polymers with different composition and/or molecular weight with significant potential for improving the sorting technology and for making it more accurate and able to separate individual waste types rather than just broad categories of similar materials.

-*pharma industries:* pharmaceutical products usually require the finest quality control and, thus, represent a huge market for novel systems.

-*space science:* applications related to space science as astrophysics can benefit from the research and development of the novel detectors and electronics developed in this project since they can be used in spectroscopic X-ray telescopes for space observatories.

-*nuclear medicine*: the advancement of knowledge, brought by the demonstrators studied within this project, will establish a scientific and technological base for the design and development of new-generation SPECT systems. Moreover, the expected energy and timing resolution can be very appealing for the next-generation PET systems.

Impacts on Italian scientific research.

DAFNE is worldwide recognized as the **most suitable machine** to perform experiments on kaonic atoms, and a very active **international community is constantly using the results obtained by kaonic atoms experiments**, craving for new measurements to test their theoretical predictions and compare with experimental findings. This is confirmed by the 390 and 122 citations of the two SIDDHARTA papers on kaonic hydrogen dating back to 2011 and 2012 [11] [12], respectively (iNSPIRE-HEP).

The EXOTIC project will be the occasion to reinforce and strengthen the INFN's and Italy's key role in the international scientific community and to become a world leader in CZT technology.

[1] doi={10.1103/PhysRevD.98.030001}

[2] doi={10.1016/j.nuclphysbps.2010.12.071}

[3] doi={10.1103/PhysRevD.88.071104}

[4] doi={10.1051/epjconf/201612900018 }

[5] doi={10.1103/PhysRevC.106.065201}

[6] doi={10.1088/0022-3700/20/4/007}

[7] doi={10.1140/epjd/e2002-00208-x}

[8] doi= {10.1103/PhysRevC.53.79}

[9] https://www.cns.s.u-tokyo.ac.jp/~gunji/tmp/tmp2/AtomicData_NuclearData_14_485.pdf

[10] https://indico.ectstar.eu/event/182/

[11] doi={10.1016/j.physletb.2011.09.011}

[12] doi={10.1016/j.nuclphysa.2011.12.008}

[13] https://agenda.infn.it/event/25725/contributions/131346/attachments/79609/103625/Eli-Friedman-talk.pdf.

[14] https://agenda.infn.it/event/25725/contributions/131349/attachments/79624/103640/foto-Jarka.pdf

HUMAN RESOURCES

Principal Investigator

Time, in percentage terms dedicated to research activities: 50

Time, in percentage terms dedicated for other activities of the project (e.g. management, and administrative activities) (for PI > 5%): 20

	N° of units involved in the project:	Time Commitment (months/person):
Principal-Investigator:	1	18
Senior Staff:	5	168
Post Docs:		
PHD Student:		
Other Support Staff:		
Total:	6	186

Detailed description of the research activities:
The tight time schedule and the high complexity of the EXOTIC project require that each member of the team be almost 100% focused on a single task. In particular, for the first two years, many parallel activities will be ongoing, from the realization of the mechanical parts and supports, of the detectors, the laboratory tests, the MC simulations, etc.
All these activities will be propedeutic for the successful realization of all the experimental and technological goals, and each Researcher hired for EXOTIC will be strongly focused on her/his tasks. Once the experimental apparatus is ready, the whole team will be involved in the exciting adventure of the two data-taking campaigns and the extraction of the physics results.

In order to avoid any possible delay in this time schedule, the hiring procedures will start immediately after the notification to the PI of the approval and financing of the project, such as to be able to allow any hired researcher to start her/his tasks on time.
In the unwanted case of some unavoidable small delay, it will be my responsibility, as a PI, to start the most important tasks of each WP and to set the ground for the new personnel to be immediately active and on schedule once employed.

Overall, I will build up a research team composed of me (70% including 20% for management) and other 5 III Level Researchers.

In particular:

R1, to be employed at LNF for the whole project. She/He will be first involved in the preparation and the equipment of the laboratory and then, as soon as the first detector units are produced, She/He will be responsible for their spectroscopic characterization in the laboratory. Each of the 400 detectors will be characterized individually and the best 300 will be chosen.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and JPARC.
The candidates for this position will be selected on the basis of experience and know-how in X-ray detector calibrations and spectroscopic characterization.

R2, to be employed at LNF for the whole project. She/He will be first involved in the overall setup (targets, trigger, anticoincidence, support structure, shielding, detector arrangement, etc) MC simulations aiming at the best optimisation of the signal-to-background ratio both for DAFNE and JPARC. Then, She/He will exploit the developed MC codes to extract crucial information for the final measurements such as the efficiencies (geometrical, detection, etc.) needed for the delivery of the

absolute yields of the kaonic atoms' transition. Finally, her/his task will be the extraction of the kaonic atoms' transitions' energies and widths.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and JPARC.
The candidates for this position will be evaluated based on their skills in GEANT4 simulations, kaonic atoms physics and data analysis.

R3, to be employed at LNF for the whole project. She/He will be first involved in the overall setup (targets, trigger, anticoincidence, support structure, shielding, detector arrangement, etc) construction based on the outcomes of the MC simulations; she/he will be responsible for the coordination with the LNF Mechanical Planning service and the contacts with the LNF workshop or with external suppliers for the final production of all the components.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and JPARC.
The candidates for this position will be evaluated based on their skills in GEANT4 simulations and experience in realizing experimental setups.

R4, to be employed at LNF for the whole project. She/He will be first involved in the preparation of all the DAQ systems needed for the various phases of EXOTIC. From the simple one to be used for the single units characterization, to the one to be used for the overall commissioning at BTF and to the final DAQ systems for the two data-taking at DAFNE and JPARC.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and JPARC.
The candidates for this position will be evaluated based on their skills in DAQ systems realization, both from a hardware and software point of view.

R5, to be employed at LNF for the last two years of the project. She/He will be first involved in the installation, tests and data-taking at BTF, DAFNE and JPARC; then, her/his main task will be the realization of the common events reconstruction codes for both the data-taking runs. Finally, she/he will work on the extraction of the new K- mass value from the acquired data.
The candidates for this position will be evaluated based on their skills in kaonic atoms' physics and data analysis.

ETHICAL ASPECTS

Indication and explanation of relevant ethical issues of the project:

There are no relevant ethical issues to report

BUDGET PLAN

Expanditure item

Expenditure item	Year 1	Year 2	Year 3	Total
A. Principal Investigator	0	0	0	0
B. Personal Costs	211400	264250	264250	739900
Senior Staff	211400	264250	264250	739900
Post Doc				0
PHD Students				0
Other Support Staff				0
C. Instruments and equipment	0	0	0	0
D. Other Operational Costs	805000	99770	84770	989540
E. Scientific and technical consulting	500000	0	0	500000
Total not including indirect costs (€)				2229440
F. Overheads				147980
Total (€)				2377420
Additional Contribution				

Total including Additional Contribution				2377420

Budget Description For Each Item In The Table :

Host Institution

The EXOTIC project will be hosted by INFN, which is the ideal environment for its successful realization; DAFNE and the National Laboratories of Frascati (LNF) have a longstanding tradition in exotic atoms physics experiments and the EXOTIC team could also rely on the expertise and know-how of the LNF staff. Moreover, INFN has recognized expertise in managing external projects.
In the first two years of the project, the experimental setup will be designed and realized, in all its parts, at LNF, where the high quality of the technical division will ensure the optimal support to the team; in particular, the Mechanical Design and Planning Service will help in the design, production and installation of all the mechanical supports.
The detector units produced by the University of Palermo (UniPa) will be tested and characterized, with a particular focus on their response time, at the BTF facility at LNF, where pulsed electron and positron beams are available for detector testing.
In the second part of the project, the apparatus will be first installed on the DAFNE machine at LNF and then at the K1.8BR facility at J-PARC. In the first case, the data-taking schedule will be decided together with the LNF management, while at J-PARC the spokespersons of the E57 experiment, which will host EXOTIC, have shown strong support for the project which could rely there also on the expertise and know-how of the J-PARC staff.

Personnel

I commit 70% (including 20% for management) of my work time to managing the project, supervising the team, and coordinating the whole set of activities. The EXOTIC project will also exploit a collaboration with “Università degli Studi di Palermo” (UniPa). This partnership is a fundamental ingredient for the success of EXOTIC, since they can count on the main experts in the digital readout of CZT detectors, protagonists of the pioneering research towards the excellent resolutions and efficiencies obtained in collaboration with IMEM-CNR of Parma.

To realize the project, I will build up and coordinate a research team composed of myself and 5 other III Level Researchers (from R1 to R5).
In particular:

R1, to be employed at LNF for the whole project. She/He will be first involved in the preparation and the equipment of the laboratory and then, as soon as the first detector units are produced. She/He will be responsible for their spectroscopic characterization in the laboratory. Each of the 400 detectors will be characterized individually and the best 300 will be chosen.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and J-PARC.
The candidates for this position will be selected based on experience and know-how in X-ray detector calibrations and spectroscopic characterization.

R2, to be employed at LNF for the whole project. She/He will be first involved in the overall setup (targets, trigger, anticoincidence, support structure, shielding, detector arrangement, etc) MC simulations aiming at the best optimisation of the signal-to-background ratio both for DAFNE and J-PARC. Then, She/He will exploit the developed MC codes to extract crucial information for the final measurements such as the efficiencies (geometrical, detection, etc.) needed for the delivery of the absolute yields of the kaonic atoms’ transition. Finally, her/his task will be the extraction of the kaonic atoms’ transitions’ energies and widths.
In the last part of the project, She/He will be also involved in the installation, tests, and data-taking at BTF, DAFNE and J-PARC.
The candidates for this position will be evaluated based on their skills in GEANT4 simulations, kaonic atoms physics and data analysis.

R3, to be employed at LNF for the whole project. She/He will be first involved in the overall setup (targets, trigger, anticoincidence, support structure, shielding, detector arrangement, etc) construction based on the outcomes of the MC simulations; she/he will be responsible for the coordination with the LNF Mechanical Planning service and the contacts with the LNF workshop or with external suppliers for the final production of all the components.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and J-PARC.
The candidates for this position will be evaluated based on their skills in GEANT4 simulations and experience in realizing experimental setups.

R4, to be employed at LNF for the whole project. She/He will be first involved in the preparation of all the DAQ systems needed for the various phases of EXOTIC. From the simple one to be used for the single unit characterization, to the one to be used for the overall commissioning at BTF and to the final DAQ systems for the two data-taking at DA NE and J-PARC.
In the last part of the project, She/He will be also involved in the installation, tests and data-taking at BTF, DAFNE and J-PARC.
The candidates for this position will be evaluated based on their skills in DAQ systems realization, both from a hardware and

software point of view.

R5, to be employed at LNF for the last two years of the project. She/He will be first involved in the installation, tests and data-taking at BTF, DA NE and J-PARC; then, her/his main task will be the realization of the common events reconstruction codes for both the data-taking runs. Finally, she/he will work on the extraction of the new K- mass value from the acquired data.
The candidates for this position will be evaluated based on their skills in kaonic atoms' physics and data analysis.

The tight time schedule and the high complexity of the EXOTIC project require that each member of the team be almost 100% focused on a single task. In particular, for the first two years many parallel activities will be ongoing, from the realization of the mechanical parts and supports, of the detectors, the laboratory tests, the MC simulations, etc.
All these activities will be propaedeutic for the successful realization of all the experimental and technological goals, and each Researcher hired for EXOTIC will be strongly focused on her/his tasks. Once the experimental apparatus is ready, the whole team will be involved in the exciting adventure of the two data-taking campaigns and of the extraction of the physics results.

The hiring procedures will start immediately after the notification to the PI of the approval and financing of the project, such as to be able to allow any hired researcher to start her/his tasks on time.
In the unwanted case of some unavoidable small delay, it will be my responsibility, as a PI, to start the most important tasks of each WP and to set the ground for the new personnel to be immediately active and on schedule once employed.

Work Packages and financial requests

In this section, I provide a detailed description, for each WP, of the related tasks (T) including their Main Responsible (MR), as well as the corresponding Milestones (M). For each WP, the financial re-quests needed to accomplish the individual tasks are indicated.

WP1 – Management and coordination. Months 1-36.

-T1.a: Personnel Recruitment. At the very early stage of the project, I will start the procedure for hiring the 5 Researchers to work 100% on EXOTIC. MR: PI
-T2.b: Research contract. At the very beginning of the project, I will start the research contract with UniPa for the realization of the front-end and readout electronics. MR: PI.
-T1.c Interaction with facilities. At the very beginning of EXOTIC, I will start interacting with the BTF team at LNF for the organization of the beam test (T5.b), as well as with the LNF Director and the E57 spokespersons to establish the data-taking schedule of EXOTIC. MR: PI.
-T1.d Supervision of all the activities. For the whole duration of the project, I will supervise, as a PI, all the EXOTIC activities. MR: PI.

Milestone(s): M1 – definition of the final data-taking schedule at DAFNE and J-PARC. This will be delivered in the framework of the WP1, T1.c and will be crucial for the continuation of the project.
Month of delivery: 15.

Financial Requests for WP1: No specific financial requests are foreseen for this WP.

WP2 – Detectors and electronics. Months 1-18.

-T2.a: Production of CZT detectors. The production of CZT detectors will be entrusted to an external producer with a tendering process where the EXOTIC-specific requests in terms of efficiency, resolution and response time will be included in the tender. This process will be done in coordination with UniPA, exploiting their excellent experience and knowledge of CZT detectors, to ensure the perfect compatibility between detectors and electronics. MR: PI-UniPa (Research Contract).
-T2.b: Production of front-end and readout electronics. This activity will be the core of the research contract with UniPa. MR: PI-UniPa (Research Contract).
-T2.c Laboratory preparation and single unit assembly. At the beginning of the project the EXOTIC laboratory and equipment will be set up, while as soon as the first detectors and electronic boards are delivered, the single-unit assembly will start. MR: R1.
-T2.d Single units characterization in the lab. As soon as the single units will be assembled, their spectroscopic characterization will be performed. MR: R1.

Milestone(s):

M2 - Characterization of all the single units in the laboratory. This will be delivered in the framework of the WP2, T2.d, and will include spectroscopic characterization and selection of the best-responding units.
Month of delivery: 18

Financial Requests for WP2:

During the T2.a, the CZT detectors will be produced. The cost is of 1800 € / cm2, and 400 will be produced in order to take into account the redundancy needed to mitigate the risks presented in the EXOTIC Summary. A total of 720000 € of “other costs” is then allocated. The front-end and readout electronics is the subject of the Research Contract to the University of Palermo and considers all the expenses they will have to face to produce all the boards (personnel, consumables, components, etc.); the total budget allocated for the Research Contract is 500000 €. For the activities T2.c and T2.d the budget of 158550 € under “personnel” is allocated for the 3-year contract of Researcher 1 (R1). R1 will work for 18 months on her/his specific tasks in this WP and then will work with the whole team for the BTF tests, the two data-taking runs, the data analysis and the dissemination activities. 35000 € of “other costs” are allocated for the cables and connectors needed for all the single-unit readouts and tests.

WP3 – MC simulations. Months 1-21.

-T3.a: Geometry and target optimization for DAFNE. This task is aimed at the optimization of all the components of the EXOTIC setup at DAFNE through MC simulations. The goal is to maximize the signal collection efficiency and the signal-to-background ratio. MR: R2.
-T3.b: Geometry and target optimization for J-PARC. This task is aimed at the optimization of all the components of the EXOTIC setup at J-PARC through MC simulations. The goal is to maximize the signal collection efficiency and the signal-to-background ratio. MR: R2.
-T3.c Efficiency estimations for final data analysis and expected results. This task is focused on the realization of the MC simulations propaedeutic to the physics results extraction, like the determination of the experimental efficiency to extract the absolute yields of the transitions. MR: R2

Milestone(s): There are no milestones foreseen within this WP

Financial Requests for WP3:

For the activities T3.a, T3.b and T3. c the budget of 158550 € under “personnel” is allocated for the 3-year contract of Researcher 2 (R2). R2 will work for 21 months on her/his specific tasks in this WP and then will work with the whole team for the two data-taking runs, for the data analysis and for the dissemination activities. From month 25, she/he will be also responsible for the extraction of the kaonic atoms ‘transitions’ energies and widths.

WP4 – Mechanics and setup. Months 1-18.

-T4.a: Realization of the box and the structure for the calibration source. This task is independent of the MC simulations and can be started at the beginning of the project. MR: R3.
-T4.b: Realization of the cooling system. The design and preparation of the cooling system are also independent of MC and will follow T4.a. MR: R3.
-T4.c Realization of the targets, trigger and coincidence. This task will start on Month 7 following T3.a and T3.b. MR: R3
-T4.d Realization of the supports for CZT. This task will start following the delivery of all the detector units (T2.c). MR: R3

Milestone(s): There are no milestones foreseen within this WP

Financial Requests for WP4:

For all the activities of WP4 the budget of 158550 € under “personnel” is allocated for the 3-year contract of Researcher 3 (R3). R3 will work for 18 months on her/his specific tasks in this WP and then will work with the whole team for the BTF tests, the two data-taking runs, the data analysis and for the dissemination activities.
For T4. a, 15000 € of “other costs” are allocated for the realization of the calibration system, including the radioactive source, the sealed box, the motorized controllers, and the carrier. For T4.b, 20000 € of “other costs” are allocated for the temperature stabilization (water cooling), including the chillers and the pipes.
For T4.c, 5000 € are allocated under “other costs” for the realization of the trigger and anticoincidence systems, 20000 € for the power supplies and the readout boards, and 5000 € for cables and connectors. Finally, 5000 € for the realization of the C, Al and S targets are also allocated under “other costs”. For T4.d, 20000 € are allocated under “other costs” for the realization of the structures to support the detector units (both at DAFNE and J-PARC) and for the shielding walls to be used at DAFNE.

WP5 – Commissioning and data taking. Months 1-33.

-T5.a: Development of the DAQ for timing and test characterization. Based on the experience gained during the SIDDHARTA-2 preliminary tests, the development of the DAQ for the CZT readout and timing test at BTF will start at the beginning of the project. MR: R4.
-T5.b: Development of the DAQ for DAFNE. The design of this DAQ will need to take care of the characteristic timing of the DAFNE machine (RF, Trigger, etc). MR: R4.
-T5.c Development of the DAQ for J-PARC. The design of this DAQ will need to take care of the characteristic timing of the K1.8BR facility and of the existing E57 components. MR: R4.
-T5.d Characterization of all the units at BTF. This task will represent one of the milestones of the project and is aimed at testing the time response of the detectors and their background rejection capabilities. MR: R4. M3.
-T5.e Installation and data taking at DAFNE. The whole EXOTIC team will be involved in this task for the setup installation

and the data-taking shifts. MR: WT.
-T5.f Installation and data taking at J-PARC. The whole EXOTIC team will be involved in this task for the setup installation and the data-taking shifts. The end of both the data taking represents a milestone for the project. MR: WT.

Milestone(s):

M3 – Characterization of the whole system at the BTF facility of LNF. This will be delivered in the framework of the WP5, T5.d, and will be mostly devoted to the characterization of the time response.
Month of delivery: 21

M4 – End of the data-taking activities. This will be delivered in the framework of the WP5, T5.f, and will consist in the successful finish of the two data-taking runs at DAFNE and J-PARC.
Month of delivery: 33

Financial Requests for WP5:

For all the activities of WP5 the budget of 158550 € under “personnel” is allocated for the 3-year contract of Researcher 4 (R4). R4 will work for 18 months on her/his specific tasks in this WP and then will work with the whole team for the BTF tests, the two data-taking runs, the data analysis and the dissemination activities.
For this WP, 40000 € of “other costs” are allocated for the mission to J-PARC, including the travel for 6 people (5000 € x 6) and the transportation of the setup (10000 €). For T5.e and T5.f , 10000 € of “other costs” are allocated for the cables and connectors to be used in DAFNE and at J-PARC.

WP6 – Data analysis and dissemination. Months 1-36.

-T6.a: Preparation of the website and of the social channels. This task is meant to set the ground for all the dissemination activities of EXOTIC. MR: PI.
-T6.b: Preparation and submission of papers. The whole team will prepare and submit papers during the entire project, publishing the outcomes both of technical research (detectors, simulations, etc) and of the physics results (kaon mass, energies, widths). MR: WT.
-T6.c Dissemination of the results at international conferences. The whole team will participate in the entire project at various international conferences presenting the status of EXOTIC and the results. MR: WT.
-T6.d Events selection and reconstruction. This part of the data analysis is devoted to particle identification, detector calibration and background reduction. MR: R5.
-T6.e Extraction of the Kaon mass value. This task will follow T6.d and will use both DAFNE and J-PARC data. MR: R5.
-T6.f Extraction of the shifts, widths and yields. This task will follow T3.c and T6.d and will use both DAFNE and J-PARC data. The completion of T6.e and T6.f is the last milestone of EXOTIC. MR: R2.
-T6.g Organization of the final workshop. This task will follow T3.c and T6.d and will use both DAFNE and J-PARC data. MR: R2. At the end of the project, an international workshop will be organized to present the final results. MR: PI.

Milestone(s):

M5 – Delivery of the scientific results of EXOTIC. This will be delivered in the framework of the WP6, T6.f, and will consist of the delivery of the new K- mass value and of the energies and widths of the KC, KAl, and KS transitions, objective of the project.
Month of delivery: 36

Financial Requests for WP6:

For all the activities of WP6 the budget of 105700 € under “personnel” is allocated for the 3-year contract of Researcher 5 (R5). R5 will work for 1 year on the realization of the event reconstruction’s code and for the second year on the extraction of the new K- mass value. In parallel, she/he will work with the whole team for the two data-taking runs and for the dissemination activities.
For T6.a, 5000 € are allocated under “other costs”, while an additional 15000 € is allocated for the organization of the final workshop.

Across all the WP, 51000 € are allocated under “other costs” for the travels of the team members outside of Italy. These are estimated to be 3000 € / year/person (5 persons for 3 years, 1 person for 2 years) and will be used both for participation in international conferences and for visits at J-PARC for the data-taking organization.

Not included in the WP tasks, 23540 € are included under “other costs” for the AUDIT expenses, estimated as 1% of the total budget. The cost is divided between the second and third years of the project. It has been assigned to milestone M5 at the closure of the project.

Budget Summary

The total cost of the EXOTIC project is 2353880 €, divided into Personnel (739900 €), Other Costs (966000 €), Research Contract (500000 €) and General Expenses (2353880 €).

In details:

Personnel is estimated from the INFN III Level Researcher contracts' cost of 52800 € / year.
EXOTIC will hire 4 Researchers for 3 years and 1 Researcher for 2 years.

Other costs are allocated for the realization of the CZT detectors (720000 €), Water cooling system (20000 €), Shielding and mechanical supports (20000 €), Power supply and readout boards for trigger and anticoincidence (20000 €), Trigger scintillators and detectors (5000 €), Calibration source and mechanics (15000 €), Cables and connectors (50000 €), Targets (5000 €), Travels outside from Italy (91000 €), and dissemination activities (20000 €).

The Research Contract (500000 €) at the University of Palermo includes all the expenses they will have to face to produce all the boards (personnel, consumables, components, etc.).

General Expenses are calculated as 20% of the total budget.

Risk assessment and mitigation: the EXOTIC contingency plan

EXOTIC is a very ambitious project aiming to pursue both scientific and technological objectives, thus facing several risks.

Risk 1: The first risk is related to a possible mismatch between the effective timing of the project, depending on its starting date, and the availability of the DAFNE and J-PARC facilities. This would result in difficulty in the organization and the time plan of the various WP and tasks. In the case of DAFNE, the SIDDHARTA-2 experiment is planned to take data until the end of 2024; the schedule after that is still to be set. The E57 experiment at J-PARC is already approved for 2027, but the exact period is unknown.
Mitigation: to mitigate the risk, the T1.c task is explicitly devoted to prompt interaction with all the facilities since the first months.
Contingency plan: in the unwanted case that one of the two measurements will result unfeasible, the statistics collected either at DAFNE or at J-PARC are sufficient to fulfil the goals of the EXOTIC project.
Risk 2: The second high risk could arise when addressing the technological challenge to produce detectors beyond their present state of the art in terms of time response, with the additional requirement of an active area of at least 1 cm2 to avoid an explosion of the number of readout channels when 300 cm2 are to be implemented. This task requires the simultaneous success of several operations, namely the design of their optimal geometry and dimensions, the production of high-quality CdZnTe crystals, the realization and bonding of the chip and the final integration with an ASIC like that developed in [28] which, however, has only been tested coupled to small devices until now.
Mitigation: to mitigate this high risk, the CdZnTe detectors will be first produced in a small amount; if in this production run a practical impossibility to proceed with this solution becomes evident, the decision to fall back to a safe and known standard geometry of CdZnTe detectors may be.
Contingency plan: In the worst-case scenario in which factor 10 is lost in S/B, the high statistics and energy resolution will ensure the high quality of the results and the fulfilment of the scientific objectives. This is clearly visible from the table in Fig. X.
Risk 3: the last risk is also related to the production of cutting-edge technology CdZnTe detector. Once the possibility of realizing 1 cm2 detectors with the originally planned dynamic range and resolution is established, the double risk of having either some failure in the bulk crystal production or a wide nonuniformity in the performances of each of them may arise. This will turn into a drop in the number of excellent detectors to be implemented in EXOTIC.
Mitigation: this risk will be mitigated by an excess in the total number of produced detectors with respect to the originally planned 300 cm2, and detailed spectroscopic characterization of each of them will be performed as foreseen in the T2.d task. In this way, the most uniform and best-performing devices will be selected for the final installation on EXOTIC. Also, if more than the originally planned 240 detectors are labelled as excellent, a redundancy in the number of ASICs and digital readout channels is foreseen so that a larger number of detectors will be installed and exploited.
Contingency plan: in the unwanted case that large nonuniformities will still be present among the best detectors, the results of the individual spectroscopic characterization will be used to develop data analysis techniques to include this effect and to ensure the high quality of the results and the fulfilment of the scientific objectives.

DECLARATION AND DOCUMENTATION

Ethics self-assessment:

Allegato 5 - Ethics Statement - EXOTIC - Signed.pdf

Commitment Letter:

lettera commitment INFN FIS 2_Scordo-signed.pdf

I hereby declare I have read the information on the personal data processing policy, pursuant to EU Regulation 2016/679 and Legislative Decree no. 196/2003 and subsequent amendments for the purposes of managing this procedure, as well as to comply with the law within the scope and for the purposes set out in the procedure

I have read the information on the personal data processing policy

YES

Upload self-certification relating to the role of main author in publications:

Dichiarazione-autore-principale-EXOTIC-Signed.pdf

Date 07/11/2023